Check for updates

FULL PAPER DOI: 10.1002/adsc.201901284

Advanced Synthesis & Catalysis

Palladium-Catalyzed Selective Three-Component Tandem Reaction to Bicyclic 1,2,3-Triazole Derivatives

Xinyu Duan,[a] Xin Huang,[a] Chunling Fu,[a] and Shengming Ma[a,*]

[a] Laboratory of Molecular Recognition and Synthesis, Department of Chemistry, Zhejiang University, Hangzhou 310027, Zhejiang, People's Republic of China
E-mail: masm@sioc.ac.cn

Manuscript received: October 4, 2019; Revised manuscript received: November 20, 2019;
Version of record online: December 11, 2019

Supporting information for this article is available on the WWW under https://doi.org/10.1002/adsc.201901284

Abstract: A Pd-catalyzed highly selective tandem reaction of allenynes, aryl or alkenyl iodides, and sodium azide for the synthesis of bicyclic 1,2,3-triazole derivatives with a very broad substrate scope including mono-substituted, 1,3-disubstituted, and trisubstituted allenynes as well as different types of organic iodides has been reported. Control experiments showed that the reaction went through a cascade allene difunctionalization/Winstein allylic azide rearrangement/intramolecular azide-alkyne cycloaddition process.

Keywords: Palladium; Bicyclic 1,2,3-triazole; Allylic azide rearrangement; Intramolecular cycloaddition

Introduction

Bicyclic 1,2,3-triazole derivatives are an important class of nitrogen-containing heterocycles because of their extensive applications in medicinal chemistry (Figure 1).[1] Traditional approaches for bicyclic triazole derivatives mainly rely on the intramolecular [3 + 2]-cycloaddition of azide with alkyne.[2–5]

Nowadays, the palladium-catalyzed intramolecular coupling cyclizations of allenes bearing nucleophilic functionalities with aryl iodides has been a powerful way to construct carbocyclic or heterocyclic compounds.[6–12] However, very limited reports have been disclosed for the intermolecular reaction of allenes, aryl iodides and nucleophiles,[13,14] due to the challenge to control the regio- and stereo-selectivity. We envisioned that if the 1,5-allenyne was applied instead of the simple allene with organic halides and sodium azide, a mixture of allylicazide isomers *Z*- or *E*-**Int-1** and **Int-2** would also be produced.[15,16] After a Cu-catalyzed intramolecular azide-alkyne cycloaddition,[2–5] bicyclic 1,2,3-triazole products **P-1** with an eight-membered ring and **P-2** with a six-membered ring may be obtained from *Z*-**Int-1** and **Int-2**, respectively, with *E*-**Int-1** remaining. Here the challenge would be the control of selectivity for the exclusive formation for one of the three products **P-1**, **P-2**, and *E*-**Int-1**. Here we wish to report the highly selective formation of [4.3.0]-bicyclic products **P-2** via the mixed intermediacy of all three intermediates, *Z*- or *E*-**Int-1** and **Int-2** (Scheme 1).

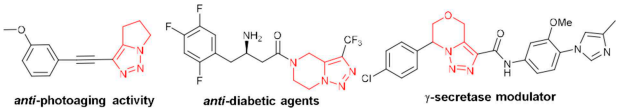
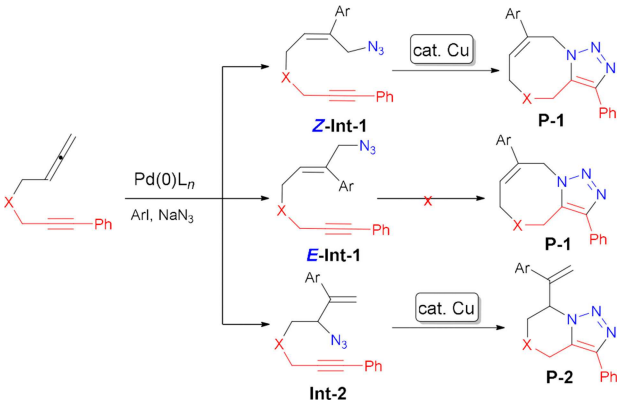


Figure 1. Some bicyclic 1,2,3-triazole derivatives with biological activities.



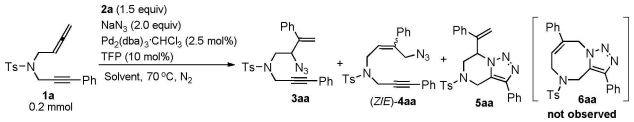
Scheme 1. The selectivity challenge in allenyne approach for bicyclic 1,2,3-triazole derivatives.

© 2019 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

Results and Discussion

With such a concept in mind, we initially chose *N*-(buta-2,3-dien-1-yl)-4-methyl-*N*-(3-phenylprop-2-ynyl) benzenesulfon-amide **1a**, iodobenzene **2a**, and NaN_3 as the model substrates. The reaction of **1a** (0.2 mmol), **2a** (1.5 equiv.), and NaN_3 (2 equiv.) in the presence of CuI (5 mol%)/$Pd_2(dba)_3 \cdot CHCl_3$ (2.5 mol%)/TFP (10 mol%) in DMF at 70 °C was conducted. Although a mixture of the allyl azide precursors **3aa** and *Z*/*E*-**4aa** were formed, **P-2**-type bicyclic triazole derivative **5aa** was formed in 48% NMR yield, exclusively.[15] None of the **P-1**-type [6.3.0]-bicyclic product **6aa** was detected (entry 1, Table 1). Encouraged by this observation, we turned to optimize the reaction conditions. Unexpectedly, the reaction afforded **5aa** in an even higher yield when CuI was removed (entry 2, Table 1). Then, various solvents were examined: the reaction in polar solvents such as DMF, NMP, DMAc (*N*,*N*-dimethylacetamide), and DMSO provided a similar yield of **5aa** (entries 2–5, Table 1). The reaction in CH_3CN produced **5aa** in a much lower yield (entry 6, Table 1). Other common solvents such as 1,4-dioxane, THF, toluene, and CH_3NO_2 all failed to give the expected products (entries 7–10, Table 1).

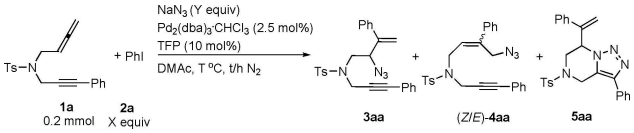
Table 1. The effects of solvents.[a]



Entry	Solvent	Time (h)	Yield of **3aa**/*Z*-**4aa**/*E*-**4aa** (%)[b]	Yield of **5aa** (%)[b]	Recovery of **1a** (%)[b]
1[c]	DMF	12	5/8/13	48	–
2	DMF	18	4/6/11	58	–
3	NMP	18	3/5/8	59	–
4	**DMAc**	**18**	**3/5/8**	**61**	–
5	DMSO	18	4/6/12	58	–
6	CH_3CN	19	6/11/20	24	16
7	Dioxane	19	–	–	88
8	THF	19	–	–	36
9	Toluene	18.5	–	–	92
10	CH_3NO_2	18.5	–	–	87

[a] Reaction conditions: **1a** (0.2 mmol), **2a** (1.5 equiv.), NaN_3 (2.0 equiv.), $Pd_2(dba)_3 \cdot CHCl_3$ (2.5 mol%), and TFP (10 mol%) in solvent (2 mL) at 70 °C under N_2 atmosphere unless otherwise noted.
[b] Determined by ^1H NMR analysis using 1,3,5-trimethylbenzene as internal standard.
[c] CuI (5 mol%) was added. TFP = Tri(2-furyl)phosphine.

Table 2. The effects of temperature and the loading amount of **2a** and NaN_3.[a]



Entry	X/Y	T (°C)	Time (h)	Yield of **3aa**/*Z*-**4aa**/*E*-**4aa** (%)[b]	Yield of **5aa** (%)[b]
1	1.5/2.0	50	18	10/19/26	15
2	1.5/2.0	70	18	3/5/8	61
3	1.5/2.0	90	6	–	67
4	1.5/2.0	100	3.5	–	72
5	1.5/2.0	120	2.5	–	70
6	1.5/2.0	140	2.4	–	60
7	**1.5/1.5**	**100**	**3.5**	–	**74**
8	2.0/2.0	100	3.5	–	70

[a] Reaction conditions: **1a** (0.2 mmol), **2a** (X equiv.), NaN_3 (Y equiv.), $Pd_2(dba)_3 \cdot CHCl_3$ (2.5 mol%), and TFP (10 mol%) in DMAc (2 mL) under N_2 atmosphere unless otherwise noted.
[b] Determined by ^1H NMR analysis using 1,3,5-trimethylbenzene as the internal standard.

Next, effects of temperature and the loadings of phenyl iodide **2a** and NaN_3 were investigated (Table 2). When the reaction was conducted at 100 °C, **5aa** was formed exclusively in 72% yield with no observation of allyl azide precursors **3aa** and **4aa** (entries 1–4, Table 2). Running the reaction at a higher temperature did not improve the yield (entries 5 and 6, Table 2). Fortunately, 74% yield of **5aa** was produced when 1.5 equiv. each of **2a** and NaN_3 were applied (entry 7, Table 2). Thus, the conditions described in entry 7 of Table 2 had been defined as the optimal conditions.

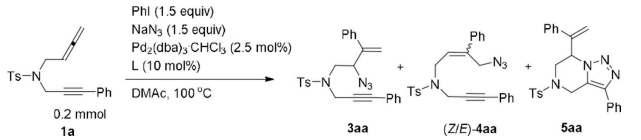
We next investigated the effect of various ligands. PPh_3 was used instead of TFP, **5aa** could be obtained just in the yield of 53% (entry 1, Table 3). When some sterically bulky monodentated phosphine ligands were used in our reaction, no product was observed (entries 2–4, Table 3). Among a set of respresentative bisphosphane ligands, the product **5aa** was obtained in 13–36% yields (entries 6–8, Table 3). Obviously, TFP was necessary in the synthesis of **5aa** (entry 9, Table 3).

The scope of organic iodides was then examined on a 0.5 mmol scale of **1a**. In addition to the phenyl group, the electron-donating groups such as methoxy and methyl substituted phenyl could be introduced to afford the desired 4,5,6,7-tetrahydro-[1,2,3]triazolo[1,5-*a*]pyrazine derivatives **5aa**–**5ad** in 75–79% yields (entries 1–4, Table 4). However, the reaction did not work with 2-methylphenyl iodide under the standard conditions. Such a method was also applicable for electron-withdrawing group substituted phenyl iodides, tolerating synthetically useful functionalities such as halogen, trifluoromethyl, ketone, and ester. The corresponding bicyclic triazole products were isolated in moderate to good yields (entries 5–11, Table 4). In

© 2019 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2020, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201901284 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

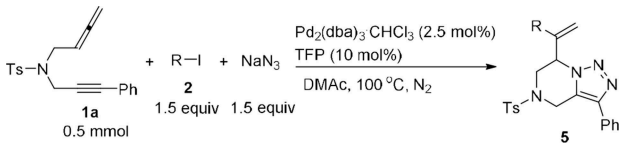
Table 3. The effects of ligands.[a]



Entry	Ligand	Time (h)	Yield of **3aa**/*Z*-**4aa**/*E*-**4aa** (%)[b]	Yield of **5aa** (%)[b]
1	PPh_3	3.5	0	53
2	Gorlos-Phos·HBF_4	17.5	0	0
3	LB-Phos·HBF_4	17.5	0	0
4	Zheda-Phos	17.5	0	0
5	XPhos	17.5	0	15
6[c]	dppp	5.5	0/3/4	13
7[c]	dppf	4.5	0	26
8[c]	DPEPhos	5.7	0	36
9	–	9.5	0	22

[a] Reaction conditions: **1a** (0.2 mmol), **2a** (1.5 equiv.), NaN_3 (1.5 equiv.), $Pd_2(dba)_3·CHCl_3$ (2.5 mol%), and ligand (10 mol%) in DMAc (2 mL) at 100 °C under N_2 atmosphere unless otherwise noted.
[b] Determined by ^1H NMR analysis using 1,3,5-trimethylbenzene as internal standard;
[c] Ligand (5 mol%) was added.

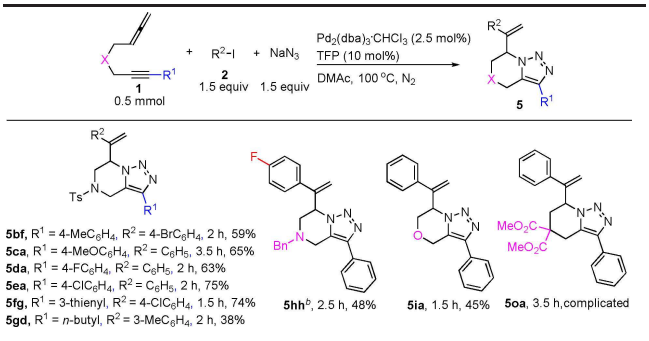
Table 4. The scope of organic iodides.[a]



Entry	**2** R	Time (h)	Yield of **5** (%)[b]
1	C_6H_5 (**2a**)	3.5	**5aa** (79)
2	4-$MeOC_6H_4$ (**2b**)	3.5	**5ab** (75)
3	4-MeC_6H_4 (**2c**)	4	**5ac** (75)
4	3-MeC_6H_4 (**2d**)	3	**5ad** (77)
5	4-IC_6H_4 (**2e**)	3	**5ae** (53)
6	4-BrC_6H_4 (**2f**)	3	**5af** (75)
7	4-ClC_6H_4 (**2g**)	3	**5ag** (74)
8	4-FC_6H_4 (**2h**)	3	**5ah** (74)
9	4-$F_3CC_6H_4$ (**2i**)	3	**5ai** (67)
10	4-$MeCOC_6H_4$ (**2j**)	2	**5aj** (68)
11	4-$MeO_2CC_6H_4$ (**2k**)	3.5	**5ak** (69)
12[c]	1-naphthalenyl (**2l**)	1.5	**5al** (57)
13	2-thienyl (**2m**)	2	**5am** (71)
14	3-pyridinyl (**2n**)	2.5	**5an** (74)
15[c,d]	(*E*)-1-hexenyl (**2o**)	2	**5ao** (40)

[a] Reaction conditions: **1a** (0.5 mmol), **2** (1.5 equiv.), NaN_3 (1.5 equiv.), $Pd_2(dba)_3·CHCl_3$ (2.5 mol%) and TFP (10 mol%) in DMAc (5 mL) at 100 °C under N_2 atmosphere unless otherwise noted.
[b] Isolated yield.
[c] The reaction was conducted at 120 °C.
[d] The reaction was conducted on a 0.2 mmol scale with **2o** (3 equiv.) in a sealed tube.

Table 5. The scope of allenynes.[a]



[a] Reaction conditions: **1** (0.5 mmol), **2** (1.5 equiv.), NaN_3 (1.5 equiv.), $Pd_2(dba)_3·CHCl_3$ (2.5 mol%) and TFP (10 mol%) in DMAc (5 mL) at 100 °C under N_2 atmosphere unless otherwise noted.
[b] The reaction was conducted on a 0.2 mmol scale.

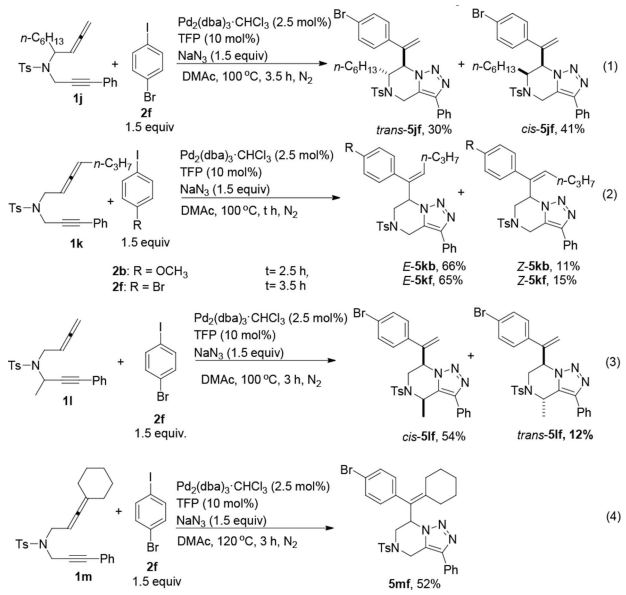
addition, 1-iodonaphthalene was also compatible, although the reaction should be conducted at 120 °C (entry 12, Table 4). Moreover, heteroaryl such as 2-thienyl and 3-pyridinyl substituted products **5am** and **5an** could be formed in 71% and 74% yields, respectively (entries 13–14, Table 4). Finally, (*E*)-1-iodohex-1-ene **2o** was also suitable for this reaction, resulting in the formation of **5ao** with a conjugated diene moiety in 40% yield (entry 15, Table 4).

Various allenynes were also tested with different phenyl iodides. R^1 group could not only be phenyl groups bearing both electron-rich and electron-deficient functionalities, but also a heteroaryl group such as 3-thienyl group. The corresponding products **5bf**–**5fg** were obtained in 59–75% yields. When R^1 group was an alkyl group such as butyl, the reaction gave bicyclic 1,2,3-triazole derivative **5gd** in a lower yield. The replacement of the tether group -NTs with -NBn or oxygen led to the formation of bicyclic products **5hh** and **5ia** in 48% and 45% yields, respectively. But when we studied the reaction of **1o**, the reaction was complicated and the product **5oa** was not formed (Table 5).

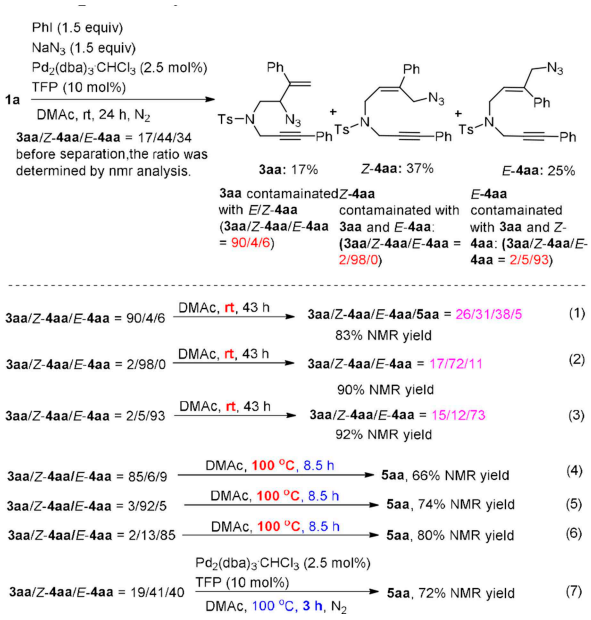
The reaction of **1j** with a *n*-hexyl group substituted at the α-position of the allene moiety delivered a pair of diastereoisomers *trans*-**5jf** and *cis*-**5jf** in 30% and 41% isolated yields, respectively (Eq. 1, Scheme 2). For the substrate **1k**, where the allene moiety is substituted with a *n*-propyl group at the terminal position, a pair of stereo-isomers were obtained (Eq. 2, Scheme 2). And the reaction of **1l** bearing a methyl substituent at the propargyl position could also be tolerated, a pair of diastereoisomers *cis*-**5lf** and *trans*-**5lf** in 54% and 12% isolated yields, respectively (Eq. 3, Scheme 2). This reaction also worked for the

© 2019 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2020, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201901284 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License



Scheme 2. The reactions of **1 j**, **1 k**, **1 l**, and **1 m**.



Scheme 3. Control experiments.

trisubstituted allenyne **1 m**, affording the product **5 mf** containing a tetra-substituted alkene moiety in 52% yield (Eq. 4, Scheme 2). The structures of *cis*-**5 jf**,[17] *E*-**5 kf**,[18] *cis*-**5 lf**[19] and **5 mf**[20] were confirmed by their X-ray single crystal diffraction analyses (Figure 2).

To gain insights into the mechanism, a series of control experiments were performed. When the reaction of **1 a** with iodobenzene in the presence of NaN_3 was conducted at room temperature, surprisingly, three allyl azides **3 aa**, *Z*-**4 aa**, and *E*-**4 aa** were obtained in the NMR yield of 17%, 44%, and 34%, respectively (Scheme 3). It was observed that these three isomers may interconvert after being stirred in DMAc at room temperature for 43 hours (Eqs. 1–3, Scheme 3).[21] In addition, heating the mixtures of allylicazide precursors **3 aa** and *Z*/*E*-**4 aa** with different ratios at 100 °C after 8.5 hours all led to the formation of **5 aa** exclusively (Eqs. 4–6, Scheme 3). As a comparison, the transformation of allylicazide precursors **3 aa** and *Z*/*E*-**4 aa** with the palladium catalyst was accomplished in only 3 hours, indicating that the palladium catalyst may accelerate the the azide-alkyne cycloaddition process (Eq. 7, Scheme 3).[22]

Based on the experimental results, a rationale was proposed in Scheme 4 by taking the reaction of **1 a** as an example. First, the oxidative addition of iodobenzene with Pd(0) formed arylpalladium **Int-3**, which would undergo insertion with the allene unit to afford (π-allyl)palladium intermediate **Int-4**. The (π-allyl) palladium species **Int-4** would accept nucleophilic attack of azide anion to provide the three allylic azide precursors **3 aa** and *Z*/*E*-**4 aa**. Finally, the intramolecu-

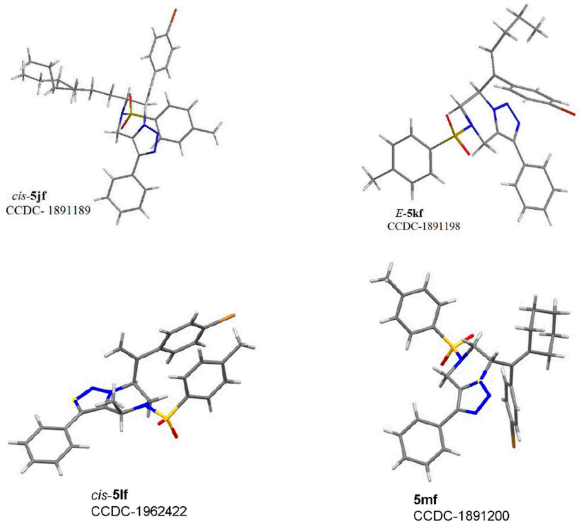
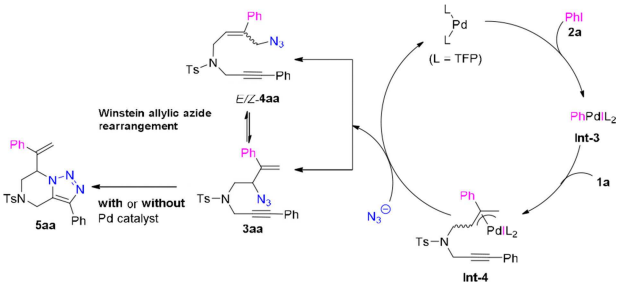


Figure 2. ORTEP representations of *cis*-**5 jf**, *E*-**5 kf**, *cis*-**5 lf**, and **5 mf**.



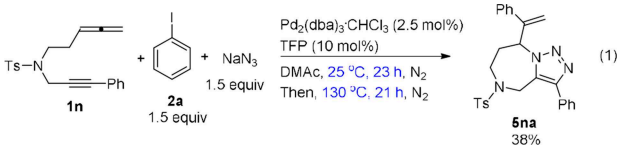
Scheme 4. The possible mechanism.

© 2019 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2020, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201901284 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

lar [3+2]-cycloaddition of **3aa** with or without Pd-catalyst produced the bicyclic 1,2,3-triazole **5aa** exclusively. *Z*/*E*-**4aa** was converted to **3aa** via Winstein allylic azide rearrangement[21] for the final exclusive conversion to **5aa**.[3e]

According to the proposed mechanism, a one-pot two-step synthesis of [5.3.0]bicyclic 1,2,3-triazole **5na** has also been realized in 38% yield (Eq. 1).



Conclusion

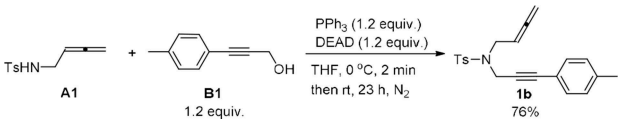
We have developed an efficient approach for the highly selective construction of [4.3.0]-bicyclic 1,2,3-triazole derivatives via a palladium-catalyzed three-component tandem reaction of allenynes, organic iodides, and NaN_3. Owing to the importance of the bicyclic 1,2,3-triazole structure as well as the good tolerance of synthetically attractive functionalities such as methoxyl, halogen, trifluoromethyl, ketone, and ester, this method would be useful in organic synthesis and medicinal chemistry. After careful mechanistic studies, a mechanism involving the interconverting between allylicazides is proposed. Further studies are being pursued in this laboratory.

Experimental Section

General Information

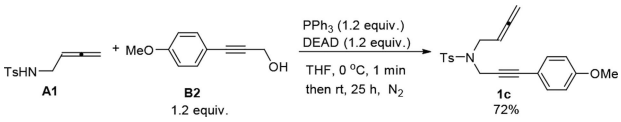
^{1}H NMR, ^{13}C NMR, and ^{19}F NMR spectra were recorded in $CDCl_3$ using a Bruker AM 300 MHz NMR spectrometer (^{1}H at 300 MHz, ^{13}C at 75 MHz, ^{19}F at 282 MHz) using TMS (^{1}H, $\delta=0$), residual $CHCl_3$ (7.26 ppm) in $CDCl_3$, and $CFCl_3$ (^{19}F $CFCl_3$, $\delta=0$) as the internal standards, respectively. IR spectra were recorded with a Perkin-Elmer 983G instrument. Elemental analyses were conducted with a Carlo-Erba EA1110 elementary analysis instrument. Mass spectrometry was performed with an HP 5989 A system. High-resolution mass spectrometry was determined with a Finnigan MAT 8430 or Bruker APEXIII instrument. $Pd_2(dba)_3\cdot CHCl_3$ was purchased from *Alfa Aesar*. TFP was purchased from *J&K*. DMAc was stirred with CaH_2 at 80 °C for 12 hours and distilled in vacuo before use. The boiling point of the petroleum ether used for chromatography ranges from 60~90 °C. The starting materials **1a**,[23] **1i**,[24] **1n**,[25] **1k**,[26] and **1m**[27] were prepared as reported in the literatures.

N-(buta-2,3-dienyl)-*N*-(3-(*p*–tolyl)prop-2-ynyl) 4-methylbenzenesulfonamide 1b



Typical Procedure I: To a dry Schlenk bottle were added PPh_3 (1.5728 g, 6 mmol), **A1** (1.1168 g, 5 mmol), and **B1** (0.8773 g, 6 mmol)/THF (15 mL) sequentially under N_2 atmosphere. The resulting mixture was cooled to 0 °C using an ice-water bath with stirring. After that, DEAD (0.95 mL, d = 1.106 g/cm^3, 1.05 g, 6 mmol) was added dropwise within 2 mins. The ice-water bath was then removed and the resulting mixture was warmed up to room temperature gradually. After being stirred for 23 h, the reaction was complete as monitored by TLC. After evaporation of the solvent, the residue was purified by column chromatography on silica gel (petroleum ether/ethyl acetate = 20/1 (525 mL) to 15/1 (480 mL)) to afford **1b** (1.3713 g, purity = 98%, 76%): oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.76 (d, J = 8.1 Hz, 2H, ArH), 7.25 (d, J = 8.1 Hz, 2H, ArH), 7.04 (d, J = 8.1 Hz, 2H, ArH), 6.98 (d, J = 8.4 Hz, 2H, ArH), 5.11 (quint, J = 6.8 Hz, 1H, =CH), 4.80 (dt, J_1 = 6.6 Hz, J_2 = 2.5 Hz, 2H, CH_2=), 4.36 (s, 2H, NCH_2), 3.92 (dt, J_1 = 7.1 Hz, J_2 = 2.3 Hz, 2H, NCH_2), 2.35 (s, 3H, CH_3), 2.32 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 209.8, 143.4, 138.5, 136.1, 131.4, 129.5, 128.8, 127.7, 119.2, 85.7, 85.6, 80.9, 76.3, 45.8, 36.8, 21.3; IR (neat) ν (cm^{-1}) 3029, 2921, 2860, 2241, 1955, 1598, 1510, 1495, 1437, 1405, 1349, 1306, 1261, 1163, 1119, 1094, 1020; MS (70 ev, EI) m/z (%) 352 (M^+ + 1, 5.01), 351 (M^+, 19.85), 193 (100); HRMS calcd for $C_{21}H_{21}NO_2S$ (M^+): 351.1293, found: 351.1293.

N-(buta-2,3-dien-1-yl)-*N*-(3-(4-methoxyphenyl) prop-2-ynyl)-4-methylbenzenesulfonamide 1c

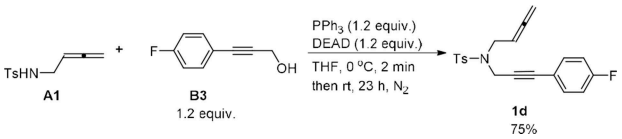


Following **Typical Procedure I**, the reaction of PPh_3 (0.3147 g, 1.2 mmol), **A1** (0.2237 g, 1 mmol), **B2** (0.1948 g, 1.2 mmol), and DEAD (0.19 mL, d = 1.106 g/cm^3, 0.21 g, 1.2 mmol) in THF (3 mL) afforded **1c** (0.2646 g, 72%) (eluent: petroleum ether/ethyl acetate = 20/1 (420 mL) to 15/1 (480 mL)): solid; m. p. 71.6–72.3 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.76 (d, J = 8.4 Hz, 2H, ArH), 7.25 (d, J = 7.8 Hz, 2H, ArH), 7.03 (d, J = 9.0 Hz, 2H, ArH), 6.76 (d, J = 9.0 Hz, 2H, ArH), 5.11 (quint, J = 6.8 Hz, 1H, CH=), 4.80 (dt, J_1 = 6.6 Hz, J_2 = 2.4 Hz, 2H, =CH_2), 4.35 (s, 2H, CH_2), 3.92 (dt, J_1 = 7.2 Hz, J_2 = 2.4 Hz, 2H, CH_2), 3.79 (s, 3H, OCH_3), 2.35 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 209.9, 159.6, 143.4, 136.1, 133.0, 129.5, 127.7, 114.3, 113.7, 85.6, 85.5, 80.2, 76.4, 55.2, 45.8, 36.9, 21.4; IR (neat) ν (cm^{-1}) 2921, 2839, 2241, 1955, 1606, 1568, 1510, 1442, 1348, 1305, 1292, 1248, 1162, 1094, 1031; MS (70 ev, EI) m/z (%) 368 (M^+ + 1, 8.77), 367 (M^+, 36.74), 185 (100); Anal. Calcd for $C_{21}H_{21}NO_3S$: C 68.64, H 5.76, N 3.81. Found: C 68.19, H 5.76, N 3.57.

© 2019 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

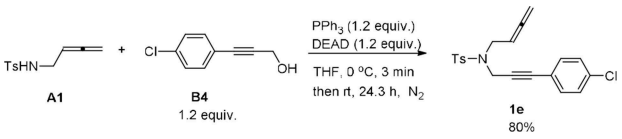
16154169, 2020, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201901284 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

N-(buta-2,3-dienyl)-*N*-(3-(4-fluorophenyl)prop-2-ynyl)-4-methylbenzenesulfonamide 1 d



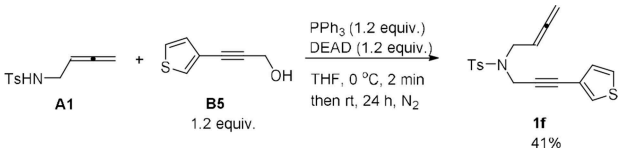
Following **Typical Procedure I**, the reaction of PPh_3 (1.5730 g, 6 mmol), **A1** (1.1173 g, 5 mmol), **B3** (0.9008 g, 6 mmol), and DEAD (0.95 mL, d = 1.106 g/cm^3, 1.05 g, 6 mmol) in THF (15 mL) afforded **1 d** (1.3329 g, 75%) (eluent: petroleum ether/ethyl acetate = 20/1 (840 mL)): oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.76 (d, J = 8.4 Hz, 2H, ArH), 7.25 (d, J = 8.1 Hz, 2H, ArH), 7.12–7.03 (m, 2H, ArH), 6.98–6.87 (m, 2H, ArH), 5.10 (quint, J = 6.9 Hz, 1H, CH=), 4.80 (dt, J_1 = 6.6 Hz, J_2 = 2.4 Hz, 2H, =CH_2), 4.35 (s, 2H, NCH_2), 3.93 (dt, J_1 = 7.1 Hz, J_2 = 2.5 Hz, 2H, NCH_2), 2.34 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 209.8, 162.4 (d, J = 248.2 Hz), 143.5, 136.0, 133.3 (d, J = 8.3 Hz), 129.4, 127.7, 118.3 (d, J = 3.5 Hz), 115.3 (d, J = 22.1 Hz), 85.4, 84.4, 81.3, 76.4, 45.8, 36.7, 21.3; ^{19}F NMR (282 MHz, $CDCl_3$) δ −111.0 (s, 1F); IR (neat) ν (cm^{-1}) 3067, 2985, 2922, 2856, 2249, 1955, 1600, 1506, 1436, 1349, 1306, 1231, 1220, 1163, 1094, 1018; MS (70 ev, EI) m/z (%) 356 (M^+ + 1, 4.49), 355 (M^+, 21.48), 198 (100); HRMS calcd for $C_{20}H_{18}NO_2SF$ (M^+): 355.1042, found: 355.1043.

N-(buta-2,3-dienyl)-*N*-(3-(4-chlorophenyl)prop-2-ynyl)-4-methylbenzenesulfonamide 1 e



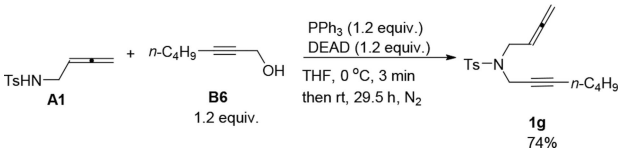
Following **Typical Procedure I**, the reaction of PPh_3 (1.5726 g, 6 mmol), **A1** (1.1170 g, 5 mmol), **B4** (1.0001 g, 6 mmol), and DEAD (0.95 mL, d = 1.106 g/cm^3, 1.05 g, 6 mmol) in THF (15 mL) afforded **1 e** (1.4875 g, 80%) (eluent: petroleum ether/ethyl acetate = 20/1 (630 mL) to 15/1 (480 mL)): solid; m.p. 75.8–76.4 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.76 (d, J = 8.1 Hz, 2H, ArH), 7.28–7.18 (m, 4H, ArH), 7.01 (d, J = 8.7 Hz, 2H, ArH), 5.10 (quint, J = 6.8 Hz, 1H, CH=), 4.80 (dt, J_1 = 6.5 Hz, J_2 = 2.5 Hz, 2H, =CH_2), 4.35 (s, 2H, NCH_2), 3.92 (dt, J_1 = 7.2 Hz, J_2 = 2.4 Hz, 2H, NCH_2), 2.35 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 209.8, 143.5, 135.9, 134.3, 132.6, 129.4, 128.4, 127.7, 120.6, 85.4, 84.4, 82.7, 76.4, 45.9, 36.7, 21.3; IR (neat) ν (cm^{-1}) 3066, 2922, 2856, 2241, 1955, 1598, 1489, 1436, 1398, 1348, 1306, 1260, 1161, 1092, 1015; MS (70 ev, EI) m/z (%) 373 (M^+ (^{37}Cl), 11.94), 371 (M^+ (^{35}Cl), 24.75), 180 (100); Anal. Calcd for $C_{20}H_{18}ClNO_2S$: C 64.60, H 4.88, N 3.77. Found: C 64.47, H 4.92, N 3.67.

N-(buta-2,3-dienyl)-*N*-(3-(thiophen-3-yl)prop-2-ynyl)-4-methyl benzenesulfonamide 1 f



Following **Typical Procedure I**, the reaction of PPh_3 (1.0403 g, 3.97 mmol), **A1** (0.7371 g, 3.3 mmol), **B5** (0.5487 g, 3.97 mmol), and DEAD (0.63 mL, d = 1.106 g/cm^3, 0.70 g, 3.97 mmol) in THF (15 mL) afforded **1 f** (0.4645 g, 41%) (eluent: petroleum ether/ethyl acetate = 20/1 (945 mL)): solid; m.p. 71.0–71.7 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.76 (d, J = 8.4 Hz, 2H, ArH), 7.26 (d, J = 8.1 Hz, 2H, ArH), 7.20 (dd, J_1 = 5.1 Hz, J_2 = 3.0 Hz, 1H, ArH), 7.14 (dd, J_1 = 2.9 Hz, J_2 = 1.1 Hz, 1H, ArH), 6.79 (dd, J_1 = 5.1 Hz, J_2 = 1.2 Hz, 1H, ArH), 5.10 (quint, J = 6.9 Hz, 1H, CH=), 4.80 (dt, J_1 = 6.6 Hz, J_2 = 2.5 Hz, 2H, =CH_2), 4.34 (s, 2H, NCH_2), 3.91 (dt, J_1 = 7.2 Hz, J_2 = 2.4 Hz, 2H, NCH_2), 2.36 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 209.8, 143.5, 135.9, 129.5, 129.4, 128.8, 127.7, 125.1, 121.2, 85.5, 81.2, 80.7, 76.4, 45.8, 36.7, 21.4; IR (neat) ν (cm^{-1}) 3109, 2920, 2864, 2241, 1955, 1598, 1495, 1436, 1348, 1305, 1162, 1094, 1018; MS (70 ev, EI) m/z (%) 344 (M^+ + 1, 4.93), 343 (M^+, 26.85), 186 (100); Anal. Calcd for $C_{18}H_{17}NO_2S_2$: C 62.95, H 4.99, N 4.08. Found: C 62.80, H 5.01, N 3.92.

N-(buta-2,3-dienyl)-*N*-(hept-2-ynyl)-4-methylbenzenesulfonamide 1 g

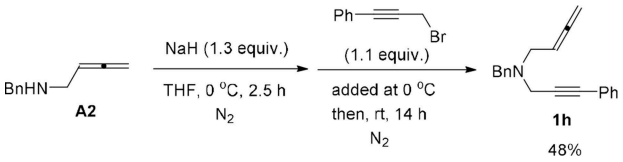


Following **Typical Procedure I**, the reaction of PPh_3 (1.5728 g, 6 mmol), **A1** (1.1170 g, 5 mmol), **B6** (0.6735 g, 6 mmol), and DEAD (0.95 mL, d = 1.106 g/cm^3, 1.05 g, 6 mmol) in THF (15 mL) afforded **1 g** (1.1904 g, purity = 98%, 74%) (eluent: petroleum ether/ethyl acetate = 20/1 (840 mL)): liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.73 (d, J = 8.1 Hz, 2H, ArH), 7.29 (d, J = 8.4 Hz, 2H, ArH), 5.05 (quint, J = 6.8 Hz, 1H, CH=), 4.77 (dt, J_1 = 6.6 Hz, J_2 = 2.4 Hz, 2H, =CH_2), 4.12 (t, J = 2.1 Hz, 2H, NCH_2), 3.85 (dt, J_1 = 7.2 Hz, J_2 = 2.5 Hz, 2H, NCH_2), 2.41 (s, 3H, CH_3), 1.99–1.82 (m, 2H, CH_2), 1.30–1.13 (m, 4H, $CH_2 \times 2$), 0.83 (t, J = 6.9 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 209.4, 143.0, 135.8, 129.0, 127.4, 85.9, 85.2, 75.9, 71.7, 45.1, 36.1, 30.0, 21.4, 21.1, 17.7, 13.1; IR (neat) ν (cm^{-1}) 2958, 2931, 2872, 2228, 1956, 1598, 1491, 1436, 1349, 1306, 1162, 1094, 1019; MS (70 ev, EI) m/z (%) 318 (M^+ + 1, 2.31), 317 (M^+, 10.76), 91 (100); HRMS calcd for $C_{18}H_{23}NO_2S$ (M^+): 317.1450, found: 317.1450.

© 2019 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

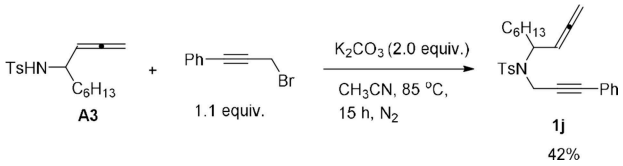
16154169, 2020, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201901284 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

N-benzyl-*N*-(3-phenylprop-2-ynyl)-*N*-buta-2,3-dienyl amine 1 h



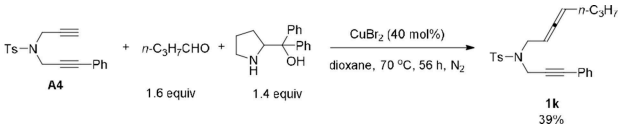
To a flame-dried Schlenk tube were added NaH (0.1042 g, 60% in mineral oil, 2.6 mmol) and THF (2 mL) under N_2 atmosphere. Then **A2** (0.3190 g, 2 mmol) was added at 0 °C (ice-water bath). After the resulting mixture was stirred at 0 °C for 2.5 hours, a solution of (3-bromoprop-1-yn-1-yl)benzene (0.4291 g, 2.2 mmol) in THF (3 mL) were added. The cooling bath was then removed. The reaction mixture was warmed up to room temperature gradually and kept stirring for 14 hours until the reaction was complete as monitored by TLC. The resulting mixture was quenched with H_2O (5 mL) and extracted with ethyl acetate (5 mL × 3). The combined organic layer was dried over anhydrous Na_2SO_4. After filtration and evaporation of the solvent, the crude residue was purified by column chromatography on silica gel (eluent: petroleum ether/ethyl ether/Et_3N = 300/5/3 (300 mL) to 300/5/6 (300 mL)) to afford **1 h** (0.2640 g, 48%): liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.54–7.20 (m, 10H, ArH), 5.23 (quint, J = 6.9 Hz, 1H, CH=), 4.76 (dt, J_1 = 6.8 Hz, J_2 = 2.6 Hz, 2H, =CH_2), 3.75 (s, 2H, NCH_2), 3.56 (s, 2H, NCH_2), 3.29 (dt, J_1 = 7.0 Hz, J_2 = 2.6 Hz, 2H, NCH_2); ^{13}C NMR (75 MHz, $CDCl_3$) δ 209.7, 138.5, 131.7, 129.2, 128.3, 128.2, 128.0, 127.1, 123.3, 87.4, 85.6, 84.4, 75.0, 57.1, 52.7, 42.3; IR (neat) ν (cm^{-1}) 3061, 3029, 2921, 2832, 1955, 1598, 1489, 1454, 1442, 1360, 1324, 1251, 1118, 1074, 1028; MS (70 ev, EI) m/z (%) 274 (M^+ + 1, 6.39), 273 (M^+, 36.23), 91 (100); HRMS calcd for $C_{20}H_{19}N$ (M^+): 273.1517, found: 273.1515.

N-(deca-1,2-dien-4-yl)-*N*-(3-phenylprop-2-yn-1-yl)-4-methyl-benzenesulfonamide 1 j



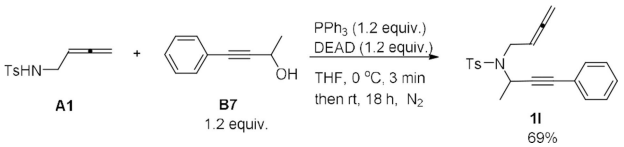
To a Schlenk tube were added K_2CO_3 (0.5530 g, 4 mmol), **A3** (0.6153 g, 2 mmol)/CH_3CN (2 mL), and (3-bromoprop-1-yn-1-yl)benzene (0.4298 g, 2.2 mmol)/CH_3CN (2 mL) sequentially under N_2 atmosphere. The resulting mixture was stirred at 85 °C for 15 h until the reaction was complete as monitored by TLC. The resulting mixture was then cooled to room temperature and filtered through a short column of silica gel (eluent: ethyl acetate (3 × 15 mL)). After evaporation of the solvent, the crude residue was purified by chromatography on silica gel (eluent: petroleum ether/ethyl acetate = 100/1 (1200 mL)) to afford **1 j** (0.3551 g, 42%): liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.82 (d, J = 8.4 Hz, 2H, ArH), 7.33–7.15 (m, 7H, ArH), 5.08 (q, J = 6.4 Hz, 1H, CH=), 4.75 (dd, J_1 = 6.3 Hz, J_2 = 3.0 Hz, 2H, =CH_2), 4.54–4.41 (m, 1H, NCH), 4.33 (d, J = 18.6 Hz, one proton of NCH_2), 4.19 (d, J = 18.6 Hz, one proton of NCH_2), 2.35 (s, 3H, CH_3), 1.78–1.56 (m, 2H, CH_2), 1.44–1.06 (m, 8H, CH_2 × 4), 0.82 (t, J = 6.8 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 208.6, 143.1, 137.9, 131.3, 129.2, 128.2, 128.1, 127.5, 122.6, 90.2, 85.3, 83.8, 77.2, 56.6, 32.9, 32.3, 31.6, 28.8, 26.3, 22.4, 21.3, 13.9; IR (neat) ν (cm^{-1}) 3062, 2927, 2857, 2241, 1954, 1598, 1490, 1439, 1339, 1305, 1162, 1092, 1055; MS (70 ev, EI) m/z (%) 421 (M^+, 1.16), 336 (100); HRMS calcd for $C_{26}H_{31}NO_2S$ (M^+): 421.2076, found: 421.2073.

N-(hepta-2,3-dien-1-yl)-4-methyl-*N*-(3-phenylprop-2-yn-1-yl) benzenesulfonamide 1 k



To a flame-dried Schlenk tube were added $CuBr_2$ (0.0897 g, 0.4 mmol), diphenyl(pyrrolidin-2-yl)methanol (0.3550 g, 1.4 mmol), **A4** (0.3230 g, 1 mmol)/dioxane (1 mL), and *n*-butyraldehyde (0.1158 g, 1.6 mmol)/dioxane (2 mL) sequentially under N_2 atmosphere. The reaction mixture was stirred at 70 °C for 56 h until **A4** was completely consumed as monitored by TLC. The resulting mixture was then cooled to room temperature, diluted with ethyl ether (5 mL), and washed with an aqueous solution of hydrochloric acid (3 M, 8 mL). The organic layer was separated and the aqueous layer was extracted with ethyl ether (5 mL × 2). The combined organic layer was washed with brine (8 mL) and dried over anhydrous Na_2SO_4. After filtration and evaporation, the residue was purified by column chromatography on silica gel (petroleum ether/ethyl acetate = 25/1 (500 mL)) to afford **1 k** (0.1483 g, 39%): oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.77 (d, J = 8.1 Hz, 2H, ArH), 7.36–7.18 (m, 5H, ArH), 7.12–6.99 (m, 2H, ArH), 5.27–5.14 (m, 1H, CH=), 5.13–4.98 (m, 1H, CH=), 4.42 (d, J = 18.3 Hz, one proton of NCH_2), 4.34 (d, J = 18.3 Hz, one proton of NCH_2), 3.95 (ddd, J_1 = 13.7 Hz, J_2 = 6.8 Hz, J_3 = 2.3 Hz, one proton of NCH_2), 3.84 (ddd, J_1 = 13.5 Hz, J_2 = 7.2 Hz, J_3 = 2.0 Hz, one proton of NCH_2), 2.33 (s, 3H, CH_3), 2.06–1.87 (m, 2H, CH_2), 1.49–1.31 (m, 2H, CH_2), 0.88 (t, J = 7.4 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 205.9, 143.4, 136.1, 131.5, 129.5, 128.3, 128.1, 127.7, 122.3, 92.6, 86.0, 85.5, 81.7, 46.5, 36.6, 30.6, 22.2, 21.4. 13.6; IR (neat) ν (cm^{-1}) 3058, 2959, 2929, 2871, 2237, 1963, 1598, 1490, 1443, 1350, 1306, 1163, 1094, 1019; MS (70 ev, EI) m/z (%) 380 (M^+ + 1, 5.57), 379 (M^+, 22.41), 91 (100); HRMS calcd for $C_{23}H_{25}NO_2S$ (M^+): 379.1606, found: 379.1608.

N-(buta-2,3-dien-1-yl)-4-methyl-*N*-(4-phenylbut-3-yn-2-yl)benzenesulfonamide (1 l)



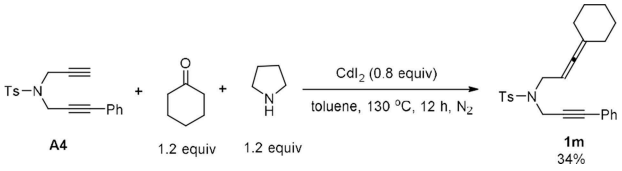
Following **Typical Procedure I**, the reaction of PPh_3 (1.5719 g, 6 mmol), **A1** (1.1170 g, 5 mmol), **B7** (0.8778 g, 6 mmol), and DEAD (0.95 mL, d = 1.106 g/cm^3, 1.05 g, 6 mmol) in THF

© 2019 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2020, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201901284 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

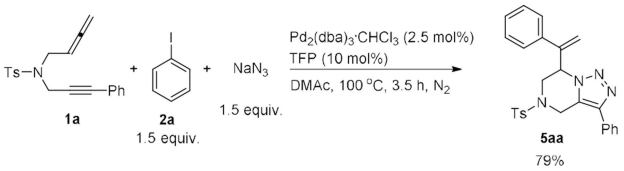
(15 mL) afforded **11** (1.2125 g, 69%) (eluent: petroleum ether/ethyl acetate = 15/1 (800 mL)): liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.78 (d, J = 8.7 Hz, 2H, ArH), 7.35–7.18 (m, 5H, ArH), 7.15–7.00 (m, 2H, ArH), 5.39–5.24 (m, 1H, CH=), 5.10 (q, J = 7.0 Hz, 1H, NCH), 4.86–4.66 (m, 2H, CH_2=), 4.07–3.94 (m, one proton of NCH_2), 3.82 (ddt, J_1 = 15.9 Hz, J_2 = 7.8 Hz, J_3 = 2.1 Hz, one proton of NCH_2), 2.35 (s, 3H, CH_3), 1.55 (d, J = 7.2 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 208.6, 143.3, 136.3, 131.4, 129.4, 128.3, 128.1, 127.6, 122.2, 89.2, 86.6, 84.9, 76.4, 46.6, 43.9, 22.4, 21.4; IR (neat) ν (cm^{-1}) 3058, 2986, 2933, 2865, 2232, 1956, 1598, 1490, 1443, 1340, 1165, 1101; MS (70 ev, EI) m/z (%) 352 (M^+ + 1, 3.34), 351 (M^+, 13.58), 180 (100); HRMS calcd for $C_{21}H_{21}NO_2S$ (M^+): 351.1293, found: 351.1292.

N-(4,4-pentamethylene-2,3-butadienyl)- *N*-(3-phenylprop-2-yn-1-yl) *p*–tosylamide 1 m



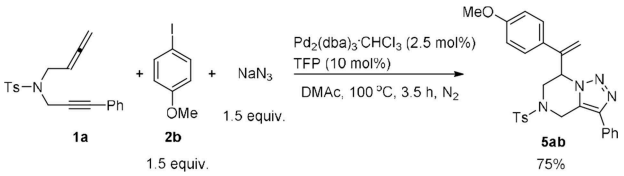
To a Schlenk tube with a polytetrafluoroethylene plug were added CdI_2 (0.8057 g, 2.2 mmol). The Schlenk tube was dried under vacuum with a heating gun until the white CdI_2 turned to yellow green and refilled with N_2 for three times. **A4** (0.8870 g, 2.7 mmol), cyclohexanone (0.34 mL, d = 0.95 g/cm^3, 0.32 g, 3.3 mmol)/toluene (4 mL), and pyrrolidine (0.28 mL, d = 0.85 g/cm^3, 0.24 g, 3.3 mmol) were added sequentially under N_2 atmosphere. The resulting mixture was stirred at 130 °C for 12 h until **A4** was completely consumed as monitored by TLC. The resulting mixture was then cooled to room temperature, diluted with ethyl acetate (10 mL), and filtered through a short column of silica gel (eluent: ethyl acetate (2 × 10 mL)). After evaporation of the solvent, the crude residue was purified by column chromatography on silica gel (petroleum ether/ethyl acetate = 25/1 (500 mL)) to afford **1 m**[28] (0.3746 g, 34%): solid; m.p. 100.7–101.4 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.77 (d, J = 8.4 Hz, 2H, ArH), 7.36–7.16 (m, 5H, ArH), 7.12–6.98 (m, 2H, ArH), 4.97–4.87 (m, 1H, CH=), 4.38 (s, 2H, CH_2), 3.86 (d, J = 7.2 Hz, 2H, CH_2), 2.32 (s, 3H, CH_3), 2.18–2.00 (m, 4H, CH_2 × 2), 1.65–1.37 (m, 6H, CH_2 × 3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 200.8, 143.3, 136.0, 131.4, 129.5, 128.2, 128.0, 127.7, 122.4, 104.1, 85.4, 83.5, 81.7, 47.0, 36.3, 31.2, 27.1, 25.9, 21.4; IR (neat) ν (cm^{-1}) 2928, 2853, 2241, 1965, 1598, 1490, 1443, 1350, 1305, 1264, 1235, 1163, 1094, 1031, 1019; MS (70 ev, EI) m/z (%) 406 (M^+ + 1, 5.97), 405 (M^+, 19.89), 91 (100).

3-Phenyl-7-(1-phenylvinyl)-5-tosyl-4,5,6,7-tetrahydro- [1,2,3]triazolo[1,5-*a*]pyrazine 5 aa



Typical Procedure II: To a flame-dried Schlenk tube were added $Pd_2(dba)_3 \cdot CHCl_3$ (0.0131 g, 0.0125 mmol), TFP (0.0117 g, 0.05 mmol), NaN_3 (0.0490 g, 0.75 mmol), **1 a** (0.1686 g, 0.5 mmol)/DMAc (2 mL), and **2 a** (0.1538 g, 0.75 mmol)/DMAc (3 mL) sequentially under N_2 atmosphere. The resulting mixture was stirred at 100 °C for 3.5 h until **1 a** was completely consumed as monitored by TLC. The resulting mixture was then cooled to room temperature, quenched with H_2O (10 mL), and extracted with ethyl acetate (15 mL × 3). The combined organic layer was washed with H_2O (10 mL) and brine (10 mL), and dried over anhydrous Na_2SO_4. After filtration and evaporation of the solvent, the crude residue was purified by column chromatography on silica gel (eluent: petroleum ether/ethyl acetate/DCM = 4/1/1 (750 mL)) to afford pure **5 aa** (0.1477 g) and impure **5 aa**, which was further purified by chromatography on silica gel (eluent: petroleum ether/ethyl acetate/DCM = 4/1/1 (450 mL)) to afford pure **5 aa** (0.0321 g). Thus, pure **5 aa** (0.1798 g, 79%) was obtained in total: solid; m.p. 139.2–140.7 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.67 (t, J = 8.0 Hz, 4H, ArH), 7.49 (t, J = 7.7 Hz, 2H, ArH), 7.44–7.28 (m, 8H, ArH), 5.70 (t, J = 3.3 Hz, 1H, CH), 5.45 (s, one proton of =CH_2), 4.92 (d, J = 15.0 Hz, one proton of NCH_2), 4.58 (s, one proton of =CH_2), 4.30 (d, J = 15.3 Hz, one proton of NCH_2), 3.77 (dd, J_1 = 12.6 Hz, J_2 = 3.5 Hz, one proton of NCH_2), 3.29 (dd, J_1 = 12.6 Hz, J_2 = 4.1 Hz, one proton of NCH_2), 2.43 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 144.6, 144.4, 141.7, 138.0, 133.0, 130.3, 130.1, 129.0, 128.7, 128.5, 128.2, 127.5, 127.2, 126.2, 125.8, 117.5, 59.6, 46.4, 42.5, 21.5; IR (KBr) ν (cm^{-1}) 3057, 2929, 1597, 1493, 1445, 1400, 1354, 1305, 1229, 1172, 1160, 1091, 1073, 1005; MS (70 ev, EI) m/z (%) 457 (M^+ + 1, 11.17), 456 (M^+, 36.13), 144 (100); Anal. Calcd for $C_{26}H_{24}N_4O_2S$: C 68.40, H 5.30, N 12.27. Found: C 68.23, H 5.33, N 12.23.

7-(1-(4-Methoxyphenyl)vinyl)-3-phenyl-5-tosyl-4,5,6,7-tetrahydro[1,2,3]triazolo[1,5-*a*]pyrazine 5 ab



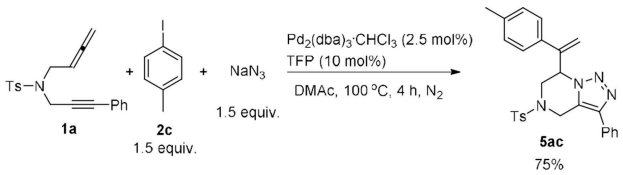
Following **Typical Procedure II**, the reaction of $Pd_2(dba)_3 \cdot CHCl_3$ (0.0128 g, 0.0125 mmol), THP (0.0118 g, 0.05 mmol), NaN_3 (0.0486 g, 0.75 mmol), **1 a** (0.1689 g, 0.5 mmol)/DMAc (2 mL), and **2 b** (0.1754 g, 0.75 mmol)/DMAc (3 mL) afford **5 ab** (0.1824 g, 75%) (eluent: petroleum ether/ethyl acetate/DCM = 4/1/1 (450 mL) to DCM (200 mL)):

© 2019 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2020, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201901284 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

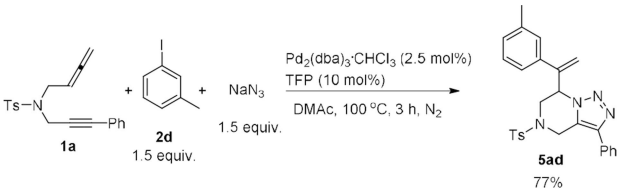
solid; m.p. 184.7–185.8 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.75–7.55 (m, 4H, ArH), 7.52–7.20 (m, 7H, ArH), 6.89 (d, J = 8.7 Hz, 2H, ArH), 5.64 (t, J = 3.5 Hz, 1H, CH), 5.36 (s, one proton of =CH_2), 4.88 (d, J = 15.3 Hz, one proton of NCH_2), 4.47 (d, J = 0.9 Hz, one proton of =CH_2), 4.28 (d, J = 15.3 Hz, one proton of NCH_2), 3.82 (s, 3H, CH_3), 3.74 (dd, J_1 = 12.6 Hz, J_2 = 3.6 Hz, one proton of NCH_2), 3.29 (dd, J_1 = 12.6 Hz, J_2 = 4.2 Hz, one proton of NCH_2), 2.42 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 159.7, 144.5, 143.8, 141.6, 133.1, 130.34, 130.29, 130.0, 128.9, 128.2, 128.1, 127.5, 126.1, 125.8, 116.1, 114.0, 59.6, 55.2, 46.4, 42.4, 21.5; IR (neat) ν (cm^{-1}) 3037, 2959, 2928, 2837, 1608, 1574, 1514, 1456, 1447, 1401, 1352, 1304, 1248, 1172, 1158, 1120, 1091, 1071, 1031, 1007; MS (70 ev, EI) m/z (%) 487 (M^+ + 1, 17.23), 486 (M^+, 56.14), 174 (100); Anal. Calcd for $C_{27}H_{26}N_4O_3S$: C 66.65, H 5.39, N 11.51. Found: C 66.37, H 5.45, N 11.37.

3-Phenyl-7-(1-(p–tolyl)vinyl)-5-tosyl-4,5,6,7-tetrahydro- [1,2,3]triazolo[1,5-*a*]pyrazine 5 ac



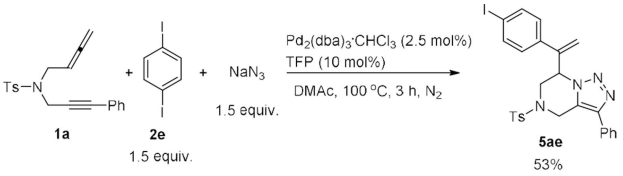
Following **Typical Procedure II**, the reaction of $Pd_2(dba)_3 \cdot CHCl_3$ (0.0127 g, 0.0125 mmol), TFP (0.0116 g, 0.05 mmol), NaN_3 (0.0485 g, 0.75 mmol), **1 a** (0.1690 g, 0.5 mmol)/DMAc (2 mL), and **2 c** (0.1638 g, 0.75 mmol)/DMAc (3 mL) afforded **5 ac** (0.1764 g, 75%) (eluent: DCM/diethyl ether = 150/1 (300 mL)): solid; m.p. 192.4–193.4 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.63 (d, J = 8.1 Hz, 4H, ArH), 7.44 (t, J = 7.5 Hz, 2H, ArH), 7.38–7.27 (m, 3H, ArH), 7.23 (d, J = 8.1 Hz, 2H, ArH), 7.16 (d, J = 8.1 Hz, 2H, ArH), 5.63 (t, J = 3.2 Hz, 1H, CH), 5.37 (s, one proton of =CH_2), 4.88 (d, J = 15.3 Hz, one proton of NCH_2), 4.46 (s, one proton of =CH_2), 4.24 (d, J = 15.3 Hz, one proton of NCH_2), 3.74 (dd, J_1 = 12.6 Hz, J_2 = 3.0 Hz, one proton of NCH_2), 3.24 (dd, J_1 = 12.9 Hz, J_2 = 4.2 Hz, one proton of NCH_2), 2.40 (s, 3H, CH_3), 2.36 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 144.4, 144.2, 141.4, 138.2, 135.0, 133.0, 130.3, 130.0, 129.3, 128.9, 128.0, 127.4, 126.8, 126.0, 125.8, 116.4, 59.4, 46.2, 42.3, 21.4, 21.0; IR (KBr) ν (cm^{-1}) 3029, 2924, 1607, 1598, 1513, 1493, 1456, 1446, 1351, 1305, 1290, 1245, 1231, 1172, 1158, 1119, 1092, 1073, 1018, 1005; MS (70 ev, EI) m/z (%) 471 (M^+ + 1, 12.55), 470 (M^+, 39.59), 158 (100); Anal. Calcd for $C_{27}H_{26}N_4O_2S$: C 68.91, H 5.57, N 11.91. Found: C 68.77, H 5.56, N 11.92.

3-Phenyl-7-(1-(*m*–tolyl)vinyl)-5-tosyl-4,5,6,7-tetrahydro- [1,2,3]triazolo[1,5-*a*]pyrazine 5 ad



Following **Typical Procedure II**, the reaction of $Pd_2(dba)_3 \cdot CHCl_3$ (0.0127 g, 0.0125 mmol), TFP (0.0114 g, 0.05 mmol), NaN_3 (0.0486 g, 0.75 mmol), **1 a** (0.1687 g, 0.5 mmol)/DMAc (2 mL), and **2 d** (0.1639 g, 0.75 mmol)/DMAc (3 mL) afforded **5 ad** (0.1812 g, 77%) (The crude residue was purified by column chromatography on silica gel (eluent: petroleum ether/ethyl acetate/DCM = 4/1/1 (450 mL)) to afford impure **5 ad** (0.1937 g), which was further purified by column chromatography on silica gel (eluent: DCM/ethyl ether = 150/1 (300 mL)) to afford pure **5 ad**): solid; m.p. 172.9–173.6 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.68–7.57 (m, 4H, ArH), 7.45 (t, J = 7.7 Hz, 2H, ArH), 7.40–7.21 (m, 4H, ArH), 7.21–7.10 (m, 3H, ArH), 5.64 (t, J = 3.2 Hz, 1H, CH), 5.39 (s, one proton of =CH_2), 4.89 (d, J = 15.0 Hz, one proton of NCH_2), 4.49 (s, one proton of =CH_2), 4.24 (d, J = 15.0 Hz, one proton of NCH_2), 3.75 (dd, J_1 = 12.9 Hz, J_2 = 3.3 Hz, one proton of NCH_2), 3.26 (dd, J_1 = 12.9 Hz, J_2 = 4.2 Hz, one proton of NCH_2), 2.41 (s, 3H, CH_3), 2.37 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 144.6, 144.5, 141.5, 138.3, 138.0, 133.0, 130.3, 130.0, 129.1, 128.9, 128.5, 128.0, 127.8, 127.4, 126.0, 125.8, 124.1, 116.9, 59.4, 46.3, 42.4, 21.5, 21.4; IR (neat) ν (cm^{-1}) 3035, 2921, 2852, 1597, 1582, 1494, 1447, 1353, 1305, 1170, 1158, 1091, 1072, 1007; MS (70 ev, EI) m/z (%) 471 (M^+ + 1, 5.06), 470 (M^+, 16.01), 91 (100); Anal. Calcd for $C_{27}H_{26}N_4O_2S$: C 68.91, H 5.57, N 11.91. Found: C 68.73, H 5.62, N 11.94.

7-(1-(4-Iodophenyl)vinyl)-3-phenyl-5-tosyl-4,5,6,7-tetrahydro- [1,2,3]triazolo[1,5-*a*]pyrazine 5 ae



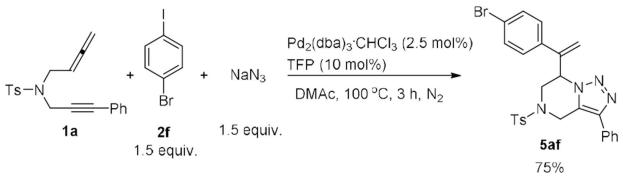
Following **Typical Procedure II**, the reaction of $Pd_2(dba)_3 \cdot CHCl_3$ (0.0127 g, 0.0125 mmol), TFP (0.0115 g, 0.05 mmol), NaN_3 (0.0486 g, 0.75 mmol), **1 a** (0.1685 g, 0.5 mmol)/DMAc (2 mL), and **2 e** (0.2606 g, 0.75 mmol)/DMAc (3 mL) afforded **5 ae** (0.1537 g, 53%) (eluent: petroleum ether/ethyl acetate/DCM = 4:1:1 (450 mL)): solid; m.p. 139.8–141.4 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.78–7.59 (m, 6H, ArH), 7.52–7.43 (m, 2H, ArH), 7.43–7.27 (m, 3H, ArH), 7.10 (d, J = 8.7 Hz, 2H, ArH), 5.63 (t, J = 3.6 Hz, 1H, CH), 5.44 (s, one proton of =CH_2), 4.91 (d, J = 15.3 Hz, one proton of NCH_2), 4.60 (d, J = 1.2 Hz, one proton of =CH_2), 4.27

© 2019 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2020, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201901284 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

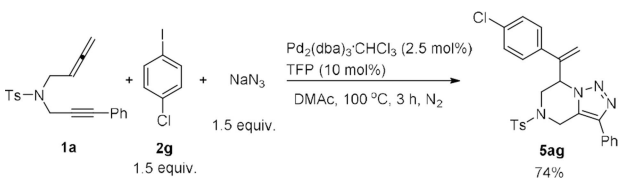
(d, J = 15.3 Hz, one proton of CH_2), 3.74 (dd, J_1 = 12.6 Hz, J_2 = 3.3 Hz, one proton of NCH_2), 3.27 (dd, J_1 = 12.8 Hz, J_2 = 4.1 Hz, one proton of CH_2), 2.43 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 144.7, 143.5, 141.7, 137.8, 137.5, 132.9, 130.2, 130.1, 129.0, 128.2, 127.5, 126.1, 125.7, 118.0, 94.3, 59.3, 46.3, 42.4, 21.5; IR (neat) ν (cm^{-1}) 3058, 2922, 2852, 1597, 1584, 1486, 1454, 1447, 1391, 1352, 1305, 1265, 1228, 1171, 1154, 1091, 1071, 1005; MS (70 ev, EI) m/z (%) 583 (M^+ + 1, 11.58), 582 (M^+, 35.80), 91(100); Anal. Calcd for $C_{26}H_{23}IN_4O_2S$: C 53.61, H 3.98, N 9.62. Found: C 53.53, H 4.28, N 9.18.

7-(1-(4-Bromophenyl)vinyl)-3-phenyl-5-tosyl-4,5,6,7-tetrahydro- [1,2,3]triazolo[1,5-*a*]pyrazine 5af



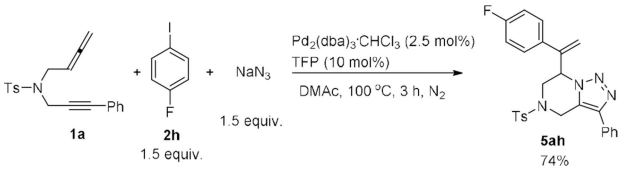
Following **Typical Procedure II**, the reaction of $Pd_2(dba)_3 \cdot CHCl_3$ (0.0129 g, 0.0125 mmol), TFP (0.0117 g, 0.05 mmol), NaN_3 (0.0490 g, 0.75 mmol), **1a** (0.1685 g, 0.5 mmol)/DMAc (2 mL), and **2f** (0.2123 g, 0.75 mmol)/DMAc (3 mL) afforded **5af** (0.2015 g, 75%) (eluent: petroleum ether/ethyl acetate/DCM = 4/1/1 (750 mL)): solid; m.p. 180.7–181.6 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.75–7.58 (m, 4H, ArH), 7.55–7.40 (m, 4H, ArH), 7.40–7.28 (m, 3H, ArH), 7.20 (d, J = 8.4 Hz, 2H, ArH), 5.59 (t, J = 3.2 Hz, 1H, CH), 5.40 (s, one proton of =CH_2), 4.90 (d, J = 15.3 Hz, one proton of NCH_2), 4.55 (s, one proton of =CH_2), 4.21 (d, J = 15.0 Hz, one proton of NCH_2), 3.72 (dd, J_1 = 12.6 Hz, J_2 = 3.0 Hz, one proton of NCH_2), 3.24 (dd, J_1 = 12.8 Hz, J_2 = 4.1 Hz, one proton of NCH_2), 2.42 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 144.6, 143.4, 141.5, 136.8, 132.9, 131.8, 130.2, 130.1, 128.9, 128.8, 128.1, 127.4, 126.0, 125.7, 122.5, 117.9, 59.2, 46.1, 42.3, 21.5; IR (neat) ν (cm^{-1}) 3063, 2955, 2922, 2852, 1597, 1489, 1447, 1394, 1352, 1305, 1260, 1229, 1171, 1091, 1072, 1007; MS (70 ev, EI) m/z (%) 536 ($M(^{81}Br)^+$, 16.88), 534 ($M(^{79}Br)^+$, 15.91), 91 (100); Anal. Calcd for $C_{26}H_{23}BrN_4O_2S$: C 58.32, H 4.33, N 10.46. Found: C 58.18, H 4.36, N 10.51.

7-(1-(4-Chlorophenyl)vinyl)-3-phenyl-5-tosyl-4,5,6,7-tetrahydro-[1,2,3]triazolo[1,5-*a*]pyrazine 5ag



Following **Typical Procedure II**, the reaction of $Pd_2(dba)_3 \cdot CHCl_3$ (0.0128 g, 0.0125 mmol), TFP (0.0117 g, 0.05 mmol), NaN_3 (0.0490 g, 0.75 mmol), **1a** (0.1688 g, 0.5 mmol)/DMAc (2 mL), and **2g** (0.1785 g, 0.75 mmol)/DMAc (3 mL) afforded **5ag** (0.1841 g, 74%) (eluent: petroleum ether/ethyl acetate/DCM = 4:1:1 (450 mL)): solid; m.p. 180.6–182.7 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.70–7.57 (m, 4H, ArH), 7.48–7.40 (m, 2H, ArH), 7.40–7.22 (m, 7H, ArH), 5.59 (t, J = 3.3 Hz, 1H, CH), 5.39 (s, one proton of =CH_2), 4.90 (d, J = 15.0 Hz, one proton of NCH_2), 4.54 (d, J = 1.2 Hz, one proton of =CH_2), 4.20 (d, J = 15.0 Hz, one proton of NCH_2), 3.72 (dd, J_1 = 12.6 Hz, J_2 = 3.0 Hz, one proton of NCH_2), 3.24 (dd, J_1 = 12.9 Hz, J_2 = 3.9 Hz, one proton of NCH_2), 2.41 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 144.6, 143.4, 141.5, 136.4, 134.3, 132.9, 130.1, 130.0, 128.9, 128.8, 128.4, 128.1, 127.4, 126.0, 125.7, 117.8, 59.3, 46.1, 42.3, 21.4; IR (neat) ν (cm^{-1}) 3055, 2959, 2924, 2860, 1597, 1493, 1456, 1447, 1398, 1352, 1306, 1266, 1228, 1171, 1158, 1091, 1072, 1007; MS (70 ev, EI) m/z (%) 492 ($M^+(^{37}Cl)$, 8.55), 490 ($M^+(^{35}Cl)$, 20.98), 91(100); Anal. Calcd for $C_{26}H_{23}ClN_4O_2S$: C 63.60, H 4.72, N 11.41. Found: C 63.24, H 4.76, N 11.29.

7-(1-(4-Fluorophenyl)vinyl)-3-phenyl-5-tosyl-4,5,6,7-tetrahydro-[1,2,3]triazolo[1,5-*a*]pyrazine 5ah

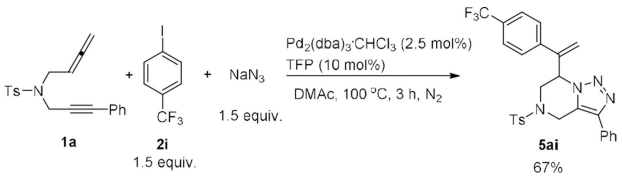


Following **Typical Procedure II**, the reaction of $Pd_2(dba)_3 \cdot CHCl_3$ (0.0128 g, 0.0125 mmol), TFP (0.0118 g, 0.05 mmol), NaN_3 (0.0490 g, 0.75 mmol), **1a** (0.1683 g, 0.5 mmol)/DMAc (2 mL), and **2h** (0.1662 g, 0.75 mmol)/DMAc (3 mL) afforded **5ah** (0.1765 g, 74%) (eluent: petroleum ether/ethyl acetate/DCM = 4:1:1 (450 mL)): solid; m.p. 158.5–159.2 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.70–7.56 (m, 4H, ArH), 7.50–7.27 (m, 7H, ArH), 7.04 (t, J = 8.6 Hz, 2H, ArH), 5.60 (s, 1H, CH), 5.37 (s, one proton of =CH_2), 4.90 (d, J = 15.3 Hz, one proton of NCH_2), 4.53 (s, one proton of =CH_2), 4.21 (d, J = 15.0 Hz, one proton of NCH_2), 3.74 (dd, J_1 = 12.9 Hz, J_2 = 3.0 Hz, one proton of CH_2), 3.25 (dd, J_1 = 12.8 Hz, J_2 = 3.8 Hz, one proton of NCH_2), 2.42 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 162.7 (d, J = 246.2 Hz), 144.6, 143.5 141.5, 134.0 (d, J = 3.5 Hz), 132.9, 130.2, 130.1, 128.934 (d, J = 8.3 Hz), 128.925, 128.1, 127.4, 126.0, 125.7, 117.5, 115.6 (d, J = 21.4 Hz), 59.5, 46.2, 42.4, 21.4; ^{19}F NMR (282 MHz, $CDCl_3$) δ −113.6 (s, 1 F); IR (neat) ν (cm^{-1}) 3063, 2959, 2925, 2852, 1602, 1511, 1456, 1447, 1401, 1353, 1306, 1233, 1173, 1160, 1092, 1072, 1007; MS (70 ev, EI) m/z (%) 475 (M^+ + 1, 5.34), 474 (M^+, 17.54), 162 (100); Anal. Calcd for $C_{26}H_{23}FN_4O_2S$: C 65.81, H 4.89, N 11.81. Found: C 66.00, H 5.04, N 11.69.

© 2019 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

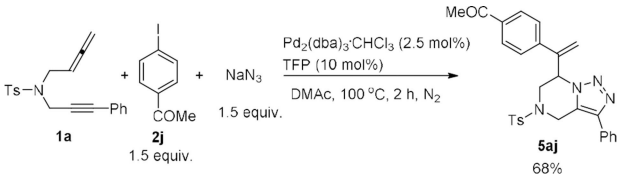
16154169, 2020, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201901284 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

3-Phenyl-5-tosyl-7-(1-(4-(trifluoromethyl)phenyl) vinyl)-4,5,6,7-tetrahydro-[1,2,3]triazolo[1,5-*a*] pyrazine 5 ai



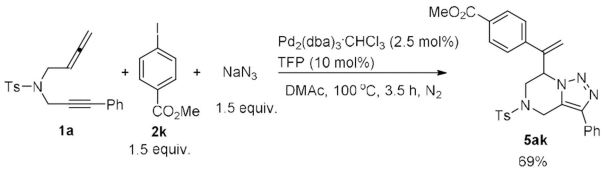
Following **Typical Procedure II**, the reaction of $Pd_2(dba)_3 \cdot CHCl_3$ (0.0127 g, 0.0125 mmol), TFP (0.0116 g, 0.05 mmol), NaN_3 (0.0487 g, 0.75 mmol), **1 a** (0.1690 g, 0.5 mmol)/DMAc (2 mL), and **2 i** (0.2043 g, 0.75 mmol)/DMAc (3 mL) afforded **5 ai** (0.1761 g, 67%) (eluent: petroleum ether/ethyl acetate/DCM = 5/1/1 (210 mL) to 4/1/1 (450 mL)): solid; m.p. 146.0–147.3 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.74–7.60 (m, 6H, ArH), 7.56–7.44 (m, 4H, ArH), 7.44–7.30 (m, 3H, ArH), 5.69 (s, 1H, CH), 5.50 (s, one proton of =CH_2), 4.95 (d, J = 15.3 Hz, one proton of NCH_2), 4.67 (s, one proton of =CH_2), 4.24 (d, J = 15.3 Hz, one proton of NCH_2), 3.75 (dd, J_1 = 12.6 Hz, J_2 = 3.0 Hz, one proton of NCH_2), 3.26 (dd, J_1 = 12.8 Hz, J_2 = 4.1 Hz, one proton of NCH_2), 2.43 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 144.8, 143.5, 141.8, 141.7, 132.9, 130.5 (q, J = 32.4 Hz), 130.2, 129.0, 128.3, 127.7, 127.5, 126.1, 125.74 (q, J = 3.7 Hz), 125.72, 123.9 (q, J = 270.5 Hz), 119.2, 59.4, 46.2, 42.5, 21.5; ^{19}F NMR (282 MHz, $CDCl_3$) δ −63.1 (s, 3F, CF_3); IR (neat) ν (cm^{-1}) 3061, 2925, 2852, 1616, 1597, 1495, 1454, 1447, 1406, 1353, 1326, 1171, 1123, 1093, 1016, 1007; MS (70 ev, EI) m/z (%) 525 (M^+ + 1, 5.42), 524 (M^+, 16.58), 91 (100); Anal. Calcd for $C_{27}H_{23}F_3N_4O_2S$: C 61.82, H 4.42, N 10.68. Found: C 61.42, H 4.43, N 10.64.

1-(4-(1-(3-Phenyl-5-tosyl-4,5,6,7-tetrahydro-[1,2,3] triazolo[1,5-*a*]pyrazin-7-yl)vinyl)phenyl)ethan-1-one 5 aj



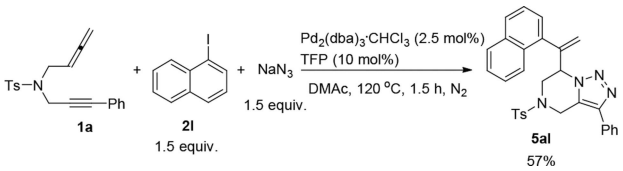
Following **Typical Procedure II**, the reaction of $Pd_2(dba)_3 \cdot CHCl_3$ (0.0127 g, 0.0125 mmol), TFP (0.0118 g, 0.05 mmol), NaN_3 (0.0486 g, 0.75 mmol), **1 a** (0.1683 g, 0.5 mmol)/DMAc (2 mL), and **2 j** (0.1842 g, 0.75 mmol)/DMAc (3 mL) afforded **5 aj** (0.1695 g, 68%) (eluent: petroleum ether/ethyl acetate/DCM = 3/1/1 (800 mL)): solid; m.p. 125.4–127.5 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 8.03–7.95 (m, 2H, ArH), 7.73–7.62 (m, 4H, ArH), 7.54–7.44 (m, 4H, ArH), 7.44–7.30 (m, 3H, ArH), 5.74 (s, 1H, CH), 5.54 (s, one proton of =CH_2), 4.94 (d, J = 15.3 Hz, one proton of NCH_2), 4.69 (s, one proton of =CH_2), 4.27 (d, J = 15.3 Hz, one proton of NCH_2), 3.76 (dd, J_1 = 12.8 Hz, J_2 = 3.5 Hz, one proton of NCH_2), 3.29 (dd, J_1 = 12.8 Hz, J_2 = 4.1 Hz, one proton of NCH_2), 2.63 (s, 3H, CH_3), 2.43 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 197.4, 144.6, 143.6, 142.5, 141.5, 136.8, 132.9, 130.1, 130.0, 128.9, 128.7, 128.1, 127.3, 127.2, 126.0, 125.7, 118.7, 59.1, 46.2, 42.4, 26.5, 21.4; IR (neat) ν (cm^{-1}) 3058, 2916, 2860, 1683, 1605, 1495, 1447, 1404, 1353, 1306, 1269, 1171, 1158, 1091, 1007; MS (70 ev, EI) m/z (%) 499 (M^+ + 1, 6.64), 498 (M^+, 22.68), 91 (100); Anal. Calcd for $C_{28}H_{26}N_4O_3S$: C 67.45, H 5.26, N 11.24. Found: C 67.33, H 5.50, N 10.87.

Methyl 4-(1-(3-phenyl-5-tosyl-4,5,6,7-tetrahydro-[1,2,3]triazolo[1,5-*a*]pyrazin-7-yl)vinyl)benzoate 5 ak



Following **Typical Procedure II**, the reaction of $Pd_2(dba)_3 \cdot CHCl_3$ (0.0129 g, 0.0125 mmol), TFP (0.0115 g, 0.05 mmol), NaN_3 (0.0486 g, 0.75 mmol), **1 a** (0.1681 g, 0.5 mmol)/DMAc (2 mL), and **2 k** (0.1962 g, 0.75 mmol)/DMAc (3 mL) afforded **5 ak** (0.1772 g, 69%) (eluent: DCM/diethyl ether = 150/1 (450 mL) to 100/1 (300 mL)): solid; m.p. 184.3–185.4 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 8.03 (d, J = 8.4 Hz, 2H, ArH), 7.64 (d, J = 8.1 Hz, 4H, ArH), 7.51–7.28 (m, 7H, ArH), 5.69 (t, J = 3.2 Hz, 1H, CH), 5.50 (s, one proton of =CH_2), 4.92 (d, J = 15.3 Hz, one proton of NCH_2), 4.63 (s, one proton of =CH_2), 4.24 (d, J = 15.0 Hz, one proton of NCH_2), 3.92 (s, 3H, OCH_3), 3.72 (dd, J_1 = 12.9 Hz, J_2 = 3.0 Hz, one proton of NCH_2), 3.28 (dd, J_1 = 12.8 Hz, J_2 = 4.1 Hz, one proton of NCH_2), 2.42 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 166.5, 144.6, 143.7, 142.4, 141.6, 132.9, 130.15, 130.06, 129.98, 129.93, 128.9, 128.1, 127.4, 127.0, 126.0, 125.7, 118.6, 59.1, 52.1, 46.2, 42.4, 21.4; IR (neat) ν (cm^{-1}) 3060, 2951, 2924, 2853, 1722, 1609, 1598, 1495, 1436, 1405, 1353, 1282, 1172, 1158, 1114, 1091, 1072, 1018, 1007; MS (70 ev, EI) m/z (%) 515 (M^+ + 1, 5.01), 514 (M^+, 13.98), 202 (100); HRMS calcd for $C_{28}H_{26}N_4O_4S$ (M^+): 514.1675, found: 514.1671.

7-(1-(naphthalen-1-yl)vinyl)-3-phenyl-5-tosyl-4,5,6,7-tetrahydro-[1,2,3]triazolo[1,5-*a*]pyrazine 5 al



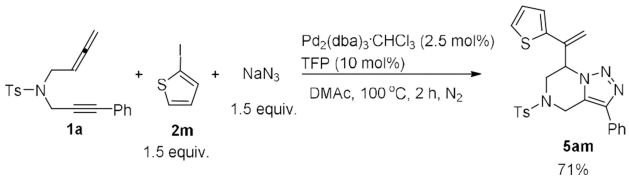
Following **Typical Procedure II**, the reaction of $Pd_2(dba)_3 \cdot CHCl_3$ (0.0134 g, 0.0125 mmol), TFP (0.0118 g, 0.05 mmol), NaN_3 (0.0487 g, 0.75 mmol), **1 a** (0.1683 g, 0.5 mmol)/DMAc (2 mL), and **2 l** (0.1903 g, 0.75 mmol)/DMAc (3 mL) afforded (0.1503 g, purity = 96%, 57%) (The crude residue was purified by column chromatography on silica gel (eluent: petroleum ether/ethyl acetate/DCM = 5/1/1 (420 mL)) to afford impure **5 al**, which was further purified by column

© 2019 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2020, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201901284 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

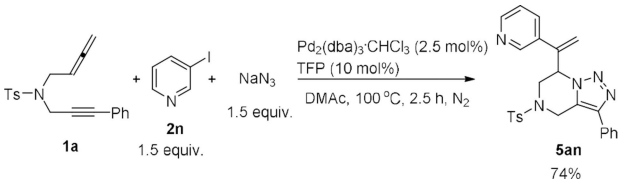
chromatography on silica gel (eluent: DCM/ethyl ether = 150/1 (320 mL)) afforded pure **5 al**): solid; m.p. 160.5–162.5 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 8.37 (d, J = 8.1 Hz, 1H, ArH), 8.00–7.80 (m, 2H, ArH), 7.73–7.40 (m, 10H, ArH), 7.40–7.27 (m, 3H, ArH), 5.58 (s, 1H, CH), 5.43 (s, one proton of =CH_2), 5.02 (d, J = 15.0 Hz, one proton of NCH_2), 4.83 (s, one proton of =CH_2), 4.11 (d, J = 15.0 Hz, one proton of NCH_2), 3.92 (dd, J_1 = 12.6 Hz, J_2 = 1.8 Hz, one proton of NCH_2), 3.02 (dd, J_1 = 12.6 Hz, J_2 = 4.2 Hz, one proton of NCH_2), 2.41 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 144.7, 143.9, 141.5, 136.0, 133.6, 132.6, 131.2, 130.3, 130.1, 128.9, 128.7, 128.4, 128.1, 127.6, 127.3, 126.9, 126.13, 126.09, 125.7, 125.5, 125.4, 119.6, 60.1, 46.0, 42.6, 21.5; IR (neat) ν (cm^{-1}) 3054, 2919, 2856, 1607, 1594, 1498, 1457, 1447, 1354, 1305, 1248, 1227, 1172, 1154, 1091, 1071, 1007; MS (70 ev, EI) m/z (%) 507 (M^+ + 1, 5.63), 506 (M^+, 14.55), 91(100); Anal. Calcd for $C_{30}H_{26}N_4O_2S$: C 71.12, H 5.17, N 11.06. Found: C 71.22, H 5.25, N 11.13.

3-Phenyl-7-(1-(thiophen-2-yl)vinyl)-5-tosyl-4,5,6,7-tetrahydro- [1,2,3]triazolo[1,5-*a*]pyrazine 5 am



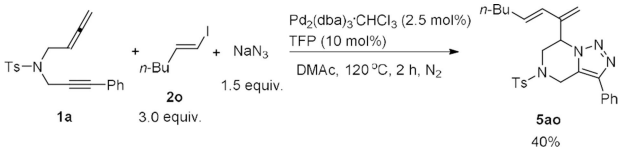
Following **Typical Procedure II**, the reaction of $Pd_2(dba)_3 \cdot CHCl_3$ (0.0130 g, 0.0125 mmol), TFP (0.0115 g, 0.05 mmol), NaN_3 (0.0490 g, 0.75 mmol), **1 a** (0.1684 g, 0.5 mmol)/DMAc (2 mL), and **2 m** (0.1576 g, 0.75 mmol)/DMAc (3 mL) afforded **5 am** (0.1646 g, 71%) (eluent: petroleum ether/ethyl acetate/DCM = 4/1/1 (750 mL)): solid; m.p. 126.6–128.0 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.71–7.58 (m, 4H, ArH), 7.51–7.41 (m, 2H, ArH), 7.40–7.33 (m, 1H, ArH), 7.30 (d, J = 8.1 Hz, 2H, ArH), 7.22 (dd, J_1 = 5.1 Hz, J_2 = 1.2 Hz, 1H, ArH), 6.96 (dd, J_1 = 5.0 Hz, J_2 = 3.8 Hz, 1H, ArH), 6.92 (dd, J_1 = 3.6 Hz, J_2 = 0.9 Hz, 1H, ArH), 5.61 (t, J = 4.4 Hz, 1H, CH), 5.57 (s, one proton of =CH_2), 4.77 (d, J = 15.3 Hz, one proton of NCH_2), 4.57 (s, one proton of =CH_2), 4.48 (d, J = 15.3 Hz, one proton of NCH_2), 3.79 (dd, J_1 = 13.1 Hz, J_2 = 4.7 Hz, one proton of NCH_2), 3.59 (dd, J_1 = 13.2 Hz, J_2 = 4.5 Hz, one proton of NCH_2), 2.40 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 144.6, 141.6, 140.1, 137.1, 133.3, 130.2, 130.0, 128.9, 128.1, 127.5, 127.4, 126.1, 126.0, 125.5, 124.2, 116.1, 59.4, 46.8, 42.3, 21.5; IR (KBr) ν (cm^{-1}) 3064, 2963, 2924, 1629, 1597, 1495, 1454, 1447, 1401, 1350, 1306, 1251, 1226, 1171, 1157, 1134, 1091, 1075, 1018, 1004; MS (70 ev, EI) m/z (%) 463 (M^+ + 1, 15.07), 462 (M^+, 51.16), 91 (100); Anal. Calcd for $C_{24}H_{22}N_4O_2S_2$: C 62.32, H 4.79, N 12.11. Found: C 62.03, H 4.87, N 12.16.

3-Phenyl-7-(1-(pyridin-3-yl)vinyl)-5-tosyl-4,5,6,7-tetrahydro- [1,2,3]triazolo[1,5-*a*]pyrazine 5 an



To a flame-dried Schlenk tube were added $Pd_2(dba)_3 \cdot CHCl_3$ (0.0130 g, 0.0125 mmol), TFP (0.0117 g, 0.05 mmol), NaN_3 (0.0490 g, 0.75 mmol), **1 a** (0.1690 g, 0.5 mmol)/DMAc (2 mL), and **2 n** (0.1587 g, 0.75 mmol)/DMAc (3 mL) sequentially under N_2 atmosphere. The reaction mixture was stirred at 100 °C for 2.5 h until **1 a** was completely consumed as monitored by TLC. The resulting mixture was then cooled to room temperature, quenched with H_2O (10 mL), and extracted with ethyl acetate (15 mL × 3). The combined organic layer was washed with H_2O (10 mL) and brine (10 mL), and dried over anhydrous Na_2SO_4. After filtration and evaporation of the solvent, the crude residue was dissolved with DCM and toluene (5 mL), followed by evaporation to get rid of DMAc. After this procedure repeated for three times, DMAc was completely removed. The crude residue was purified by column chromatography on silica gel (eluent:eluent:petroleum ether/ethyl acetate/DCM = 1/2/1 (400 mL) to 1/4/1 (600 mL)) to afford **5 an** (0.1691 g, 74%): solid; m.p. 170.8–171.7 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 8.61 (s, 2H, ArH), 7.78–7.58 (m, 5H, ArH), 7.54–7.25 (m, 6H, ArH), 5.67 (t, J = 3.5 Hz, 1H, CH), 5.52 (s, one proton of =CH_2), 4.88 (d, J = 15.3 Hz, one proton of NCH_2), 4.78 (s, one proton of =CH_2), 4.30 (d, J = 15.3 Hz, one proton of NCH_2), 3.72 (dd, J_1 = 12.8 Hz, J_2 = 3.5 Hz, one proton of NCH_2), 3.34 (dd, J_1 = 12.8 Hz, J_2 = 4.1 Hz, one proton of NCH_2), 2.43 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 149.7, 148.0, 144.8, 141.8, 141.5, 134.9, 133.8, 132.9, 130.2, 129.0, 128.2, 127.4, 126.1, 125.7, 123.5, 119.7, 59.5, 46.4, 42.5, 21.5; IR (KBr) ν (cm^{-1}) 1596, 1495, 1446, 1412, 1350, 1302, 1173, 1158, 1091, 1070, 1001; MS (70 ev, EI) m/z (%) 458 (M^+ + 1, 4.55), 457 (M^+, 15.26), 91(100); Anal. Calcd for $C_{25}H_{23}N_5O_2S$: C 65.63, H 5.07, N 15.31. Found: C 65.23, H 5.18, N 15.20.

(*E*)-7-(octa-1,3-dien-2-yl)-3-phenyl-5-tosyl-4,5,6,7-tetrahydro- [1,2,3]triazolo[1,5-*a*]pyrazine 5 ao



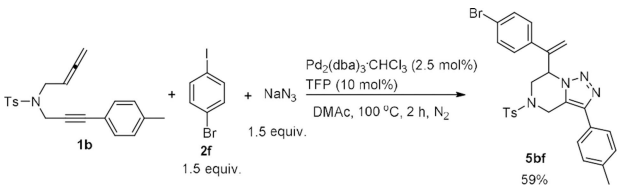
To a flame-dried Schlenk tube with a polytetrafluoroethylene plug were added $Pd_2(dba)_3 \cdot CHCl_3$ (0.0052 g, 0.005 mmol), TFP (0.0044 g, 0.02 mmol), NaN_3 (0.0194 g, 0.3 mmol), **1 a** (0.0678 g, 0.2 mmol)/DMAc (0.5 mL), and **2 o** (0.1262 g, 0.6 mmol)/DMAc (1.5 mL) sequentially under N_2 atmosphere. The Schlenk tube was then sealed by screwing the polytetra-

© 2019 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2020, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201901284 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

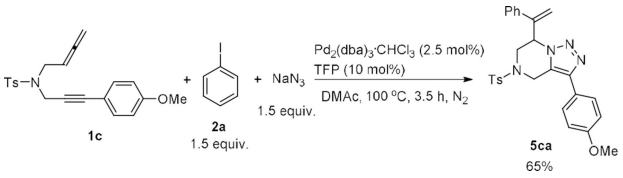
fluoroethylene plug tightly with the outlet connected to the vacuum line and the nitrogen flow being closed. The resulting mixture was stirred at 120 °C for 2 h until **1a** was completely consumed as monitored by TLC. The resulting mixture was then cooled to room temperature, quenched with H_2O (8 mL), and extracted with ethyl acetate (8 mL × 3). The combined organic layer was washed with H_2O (8 mL) and brine (8 mL), and dried over anhydrous Na_2SO_4. After filtration and evaporation of the solvent, the crude residue was purified by column chromatography on silica gel (eluent: petroleum ether/ethyl acetate/DCM = 6/1/1 (400 mL)) to afford **5ao** (0.0369 g, 40%) (eluent: petroleum ether (60 °C ~ 90 °C)/ethyl acetate/DCM = 6/1/1 (400 mL)): oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.75–7.63 (m, 4H, ArH), 7.52–7.42 (m, 2H, ArH), 7.42–7.30 (m, 3H, ArH), 5.99 (d, J = 15.9 Hz, 1H, =CH), 5.74 (dt, J_1 = 16.1 Hz, J_2 = 6.8 Hz, 1H, =CH), 5.46 (t, J = 4.4 Hz, 1H, CH), 5.18 (s, one proton of $=CH_2$), 4.77 (d, J = 15.3 Hz, one proton of NCH_2), 4.49 (d, J = 15.3 Hz, one proton of NCH_2), 4.38 (s, one proton of $=CH_2$), 3.81 (dd, J_1 = 12.9 Hz, J_2 = 4.5 Hz, one proton of NCH_2), 3.55 (dd, J_1 = 12.8 Hz, J_2 = 4.4 Hz, one proton of NCH_2), 2.44 (s, 3H, CH_3), 2.12 (q, J = 6.6 Hz, 2H, CH_2), 1.46–1.22 (m, 4H, CH_2 × 2), 0.92 (t, J = 7.2 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 144.5, 141.6, 141.0, 133.5, 132.0, 130.4, 130.0, 128.9, 128.0, 127.6, 126.1, 125.9, 116.4, 57.7, 46.9, 42.3, 32.6, 31.1, 22.1, 21.5, 13.8; IR (neat) ν (cm^{-1}) 2956, 2927, 2857, 1594, 1493, 1454, 1446, 1353, 1302, 1171, 1156, 1091, 1007; MS (70 ev, EI) m/z (%) 463 (M^+ + 1, 28.21), 462 (M^+, 92.79), 91 (100); HRMS calcd for $C_{26}H_{30}N_4O_2S$ (M^+): 462.2089, found: 462.2089.

7-(1-(4-Bromophenyl)vinyl)-3-(p–tolyl)-5-tosyl-4,5,6,7-tetrahydro-[1,2,3]triazolo[1,5-*a*]pyrazine 5bf



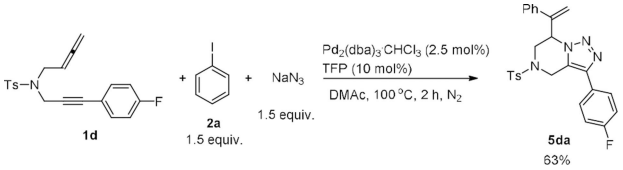
Following **Typical Procedure II**, the reaction of $Pd_2(dba)_3 \cdot CHCl_3$ (0.0129 g, 0.0125 mmol), TFP (0.0118 g, 0.05 mmol), NaN_3 (0.0490 g, 0.75 mmol), **1b** (0.1759 g, 0.5 mmol)/DMAc (2 mL), and **2f** (0.2122 g, 0.75 mmol)/DMAc (3 mL) afforded **5bf** (0.1614 g, 59%) (eluent: petroleum ether/ethyl acetate/DCM = 4/1/1 (540 mL)): solid; m.p. 176.9–177.5 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.65 (d, J = 8.1 Hz, 2H, ArH), 7.60–7.45 (m, 4H, ArH), 7.37–7.18 (m, 6H, ArH), 5.61 (t, J = 3.5 Hz, 1H, CH), 5.43 (s, one proton of $=CH_2$), 4.89 (d, J = 15.0 Hz, one proton of NCH_2), 4.60 (s, one proton of $=CH_2$), 4.24 (d, J = 15.3 Hz, one proton of NCH_2), 3.73 (dd, J_1 = 12.6 Hz, J_2 = 3.3 Hz, one proton of NCH_2), 3.26 (dd, J_1 = 12.9 Hz, J_2 = 4.2 Hz, one proton of NCH_2), 2.43 (s, 3H, CH_3), 2.41 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 144.6, 143.5, 141.7, 138.0, 136.9, 133.0, 131.8, 130.1, 129.6, 128.8, 127.4, 127.3, 126.0, 125.3, 122.5, 117.9, 59.3, 46.2, 42.4, 21.5, 21.2; IR (neat) ν (cm^{-1}) 3030, 2922, 2861, 1597, 1588, 1511, 1489, 1456, 1394, 1352, 1305, 1227, 1172, 1159, 1120, 1092, 1072, 1004; MS (70 ev, EI) m/z (%) 550 (M^+(^{81}Br), 4.55), 548 (M^+(^{79}Br), 4.13), 91 (100); Anal. Calcd for $C_{27}H_{25}BrN_4O_2S$: C 59.02, H 4.59, N 10.20. Found: C 58.98, H 4.63, N 10.24.

3-(4-Methoxyphenyl)-7-(1-phenylvinyl)-5-tosyl-4,5,6,7-tetrahydro-[1,2,3]triazolo[1,5-*a*]pyrazine 5ca



Following **Typical Procedure II**, the reaction of $Pd_2(dba)_3 \cdot CHCl_3$ (0.0127 g, 0.0125 mmol), TFP (0.0114 g, 0.05 mmol), NaN_3 (0.0489 g, 0.75 mmol), **1c** (0.1834 g, 0.5 mmol)/DMAc (2 mL), and **2a** (0.1533 g, 0.75 mmol)/DMAc (3 mL) afforded **5ca** (0.1582 g, 65%) (The crude residue was first purified by column chromatography on silica gel (eluent: petroleum ether/ethyl acetate/DCM = 4/1/1 (450 mL) to DCM/diethyl ether = 150/1 (450 mL)) to afford impure **5ca**, which was further purified by column chromatography on silica gel (eluent: DCM/diethyl ether = 150/1 (450 mL) to afford pure **5ca**): solid; m.p. 195.3–196.3 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) 7.65 (d, J = 8.1 Hz, 2H, ArH), 7.60 (d, J = 9.0 Hz, 2H, ArH), 7.45–7.28 (m, 7H, ArH), 7.00 (d, J = 9.0 Hz, 2H, ArH), 5.66 (t, J = 3.3 Hz, 1H, CH), 5.43 (s, one proton of $=CH_2$), 4.87 (d, J = 15.0 Hz, one proton of NCH_2), 4.56 (d, J = 0.9 Hz, one proton of $=CH_2$), 4.25 (d, J = 15.3 Hz, one proton of NCH_2), 3.86 (s, 3H, OMe), 3.74 (dd, J_1 = 12.8 Hz, J_2 = 3.5 Hz, one proton of NCH_2), 3.28 (dd, J_1 = 12.8 Hz, J_2 = 4.1 Hz, one proton of NCH_2), 2.42 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 159.5, 144.6, 144.4, 141.6, 138.1, 133.0, 130.1, 128.7, 128.4, 127.51, 127.47, 127.1, 124.9, 122.9, 117.4, 114.4, 59.6, 55.3, 46.4, 42.4, 21.5; IR (KBr) ν (cm^{-1}) 3094, 3058, 3009, 2958, 2930, 2833, 1616, 1598, 1570, 1509, 1494, 1460, 1444, 1378, 1350, 1304, 1292, 1255, 1228, 1173, 1157, 1092, 1074, 1036, 1015; MS (70 ev, EI) m/z (%) 487 (M^+ + 1, 12.33), 486 (M^+, 38.69), 91 (100); HRMS calcd for $C_{27}H_{26}N_4O_3S$ (M^+): 486.1726, found: 486.1722.

3-(4-Fluorophenyl)-7-(1-phenylvinyl)-5-tosyl-4,5,6,7-tetrahydro-[1,2,3]triazolo[1,5-*a*]pyrazine 5da



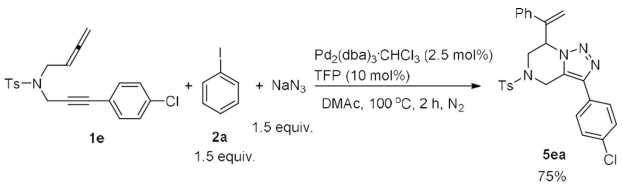
Following **Typical Procedure II**, the reaction of $Pd_2(dba)_3 \cdot CHCl_3$ (0.0051 g, 0.005 mmol), TFP (0.0047 g, 0.02 mmol), NaN_3 (0.0192 g, 0.3 mmol), **1d** (0.0708 g, 0.2 mmol)/DMAc (1 mL), and **2a** (0.0615 g, 0.3 mmol)/DMAc (1 mL) afforded **5da** (0.0597 g, 63%) (eluent: petroleum ether/ethyl acetate/DCM = 4/1/1 (300 mL)): solid; m.p. 185.8–186.3 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.73–7.62 (m, 4H, ArH), 7.45–7.30 (m, 7H, ArH), 7.24–7.13 (m, 2H,

© 2019 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2020, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201901284 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

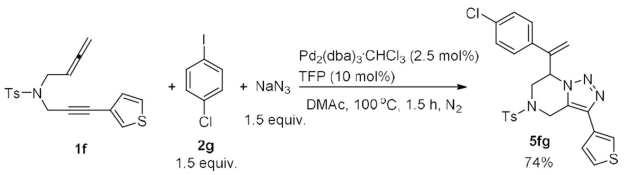
ArH), 5.71 (t, J = 3.6 Hz, 1H, CH), 5.46 (s, one proton of =CH_2), 4.88 (d, J = 15.3 Hz, one proton of CH_2), 4.60 (s, one proton of =CH_2), 4.29 (d, J = 15.3 Hz, one proton of CH_2), 3.76 (dd, J_1 = 12.9 Hz, J_2 = 3.6 Hz, one proton of CH_2), 3.31 (dd, J_1 = 12.8 Hz, J_2 = 4.1 Hz, one proton of CH_2), 2.44 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 162.4 (d, J = 246.8 Hz), 144.6, 144.4, 140.8, 138.0, 133.0, 130.1, 128.7, 128.5, 127.9 (d, J = 8.3 Hz), 127.5, 127.1, 126.5 (d, J = 2.8 Hz), 125.5, 117.4, 116.0 (d, J = 21.4 Hz), 59.6, 46.4, 42.3, 21.5; ^{19}F NMR (282 MHz, $CDCl_3$) δ −113.4 (s, 1F); IR (neat) ν (cm^{-1}) 3058, 2924, 2586, 1597, 1574, 1509, 1456, 1444, 1410, 1352, 1306, 1297, 1231, 1172, 1158, 1092, 1069, 1005; MS (70 ev, EI) m/z (%) 475 (M^+ + 1, 5.73), 474 (M^+, 18.13), 91 (100); Anal. Calcd for $C_{26}H_{23}FN_4O_2S$: C 65.81, H 4.89, N 11.81. Found: C 65.62, H 4.88, N 11.71.

3-(4-Chlorophenyl)-7-(1-phenylvinyl)-5-tosyl-4,5,6,7-tetrahydro-[1,2,3]triazolo[1,5-*a*]pyrazine 5ea



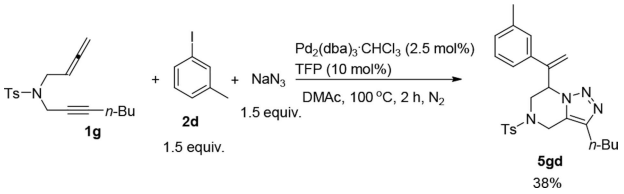
Following **Typical Procedure II**, the reaction of $Pd_2(dba)_3 \cdot CHCl_3$ (0.0127 g, 0.0125 mmol), TFP (0.0117 g, 0.05 mmol), NaN_3 (0.0485 g, 0.75 mmol), **1e** (0.1859 g, 0.5 mmol)/DMAc (2 mL), and **2a** (0.1535 g, 0.75 mmol)/DMAc (3 mL) afforded **5ea** (0.1841 g, 75%) (The crude residue was first purified by column chromatography on silica gel (eluent: petroleum ether/ethyl acetate/DCM = 5/1/1 (210 mL) to 4/1/1 (300 mL)) to afford impure **5ea**, which was further purified by chromatography on silica gel (eluent: DCM/ethyl ether = 150/1 (300 mL)) to afford pure **5ea**): solid; m.p. 182.7–183.5 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) 7.72–7.58 (m, 4H, ArH), 7.50–7.30 (m, 9H, ArH), 5.70 (t, J = 3.6 Hz, 1H, CH), 5.45 (s, one proton of =CH_2), 4.87 (d, J = 15.3 Hz, one proton of CH_2), 4.58 (d, J = 1.2 Hz, one proton of =CH_2), 4.28 (d, J = 15.3 Hz, one proton of CH_2), 3.76 (dd, J_1 = 12.9 Hz, J_2 = 3.6 Hz, one proton of CH_2), 3.30 (dd, J_1 = 12.9 Hz, J_2 = 4.2 Hz, one proton of CH_2), 2.43 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 144.7, 144.3, 140.6, 137.9, 133.9, 132.9, 130.1, 129.1, 128.8, 128.7, 128.5, 127.5, 127.3, 127.1, 125.9, 117.4, 59.6, 46.3, 42.3, 21.5; IR (neat) ν (cm^{-1}) 3058, 2916, 2856, 1597, 1494, 1456, 1405, 1352, 1304, 1229, 1171, 1159, 1092, 1003; MS (70 ev, EI) m/z (%) 492 (M^+(^{37}Cl), 6.77), 490 (M^+(^{35}Cl), 15.78), 91 (100); HRMS calcd for $C_{26}H_{23}N_4O_2SCl$ (M^+(^{35}Cl)): 490.1230, found: 490.1230.

7-(1-(4-Chlorophenyl)vinyl)-3-(thiophen-3-yl)-5-tosyl-4,5,6,7-tetrahydro-[1,2,3]triazolo[1,5-*a*]pyrazine 5fg



Following **Typical Procedure II**, the reaction of $Pd_2(dba)_3 \cdot CHCl_3$ (0.0128 g, 0.0125 mmol), TFP (0.0117 g, 0.05 mmol), NaN_3 (0.0486 g, 0.75 mmol), **1f** (0.1719 g, 0.5 mmol)/DMAc (2 mL), and **2g** (0.1787 g, 0.75 mmol)/DMAc (3 mL) afforded **5fg** (0.1840 g, 74%) (eluent: petroleum ether/ethyl acetate/DCM = 4/1/1 (750 mL)): solid; m.p. 170.8–171.2 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) 7.66 (d, J = 8.1 Hz, 2H, ArH), 7.52–7.40 (m, 3H, ArH), 7.40–7.27 (m, 6H, ArH), 5.64 (s, 1H, CH), 5.42 (s, one proton of =CH_2), 4.87 (d, J = 15.0 Hz, one proton of CH_2), 4.57 (s, one proton of =CH_2), 4.17 (d, J = 15.3 Hz, one proton of CH_2), 3.76 (dd, J_1 = 12.8 Hz, J_2 = 3.2 Hz, one proton of CH_2), 3.25 (dd, J_1 = 12.6 Hz, J_2 = 4.2 Hz, one proton of CH_2), 2.44 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 144.6, 143.3, 138.1, 136.3, 134.2, 132.7, 131.0, 130.1, 128.8, 128.4, 127.4, 126.6, 125.5, 125.1, 121.1, 117.7, 59.2, 46.1, 41.9, 21.5; IR (neat) ν (cm^{-1}) 3105, 2929, 2852, 1596, 1493, 1456, 1420, 1394, 1353, 1336, 1227, 1169, 1091, 1012; MS (70 ev, EI) m/z (%) 498 (M^+(^{37}Cl), 5.09), 496 (M^+(^{35}Cl), 10.56), 42 (100); Anal. Calcd for $C_{24}H_{21}ClN_4O_2S_2$: C 58.00, H 4.26, N 11.27. Found: C 57.92, H 4.32, N 11.28.

3-Butyl-7-(1-(*m*–tolyl)vinyl)-5-tosyl-4,5,6,7-tetrahydro- [1,2,3]triazolo[1,5-*a*]pyrazine 5gd



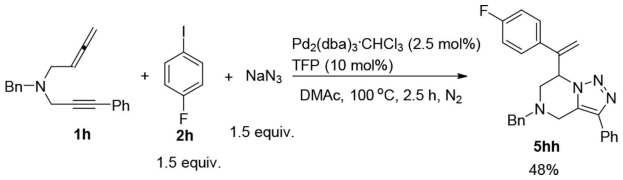
Following **Typical Procedure II**, the reaction of $Pd_2(dba)_3 \cdot CHCl_3$ (0.0131 g, 0.0125 mmol), TFP (0.0113 g, 0.05 mmol), NaN_3 (0.0490 g, 0.75 mmol), **1g** (0.1621 g, purity = 98%, 0.5 mmol)/DMAc (2 mL), and **2d** (0.1632 g, 0.75 mmol)/DMAc (3 mL) afforded **5gd** (0.0853 g, 38%) (eluent: petroleum ether/ethyl acetate/DCM = 4/1/1 (750 mL)): oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.63 (d, J = 8.1 Hz, 2H, ArH), 7.40–7.20 (m, 3H, ArH), 7.20–7.08 (m, 3H, ArH), 5.61 (t, J = 3.6 Hz, 1H, CH), 5.37 (s, one proton of =CH_2), 4.61 (d, J = 15.0 Hz, one proton of CH_2), 4.45 (s, one proton of =CH_2), 4.02 (d, J = 15.0 Hz, one proton of CH_2), 3.69 (dd, J_1 = 12.8 Hz, J_2 = 3.5 Hz, one proton of CH_2), 3.22 (dd, J_1 = 12.8 Hz, J_2 = 4.1 Hz, one proton of CH_2), 2.65 (t, J = 7.7 Hz, 2H, CH_2), 2.41 (s, 3H, CH_3), 2.36 (s, 3H, CH_3), 1.73–1.58 (m, 2H, CH_2), 1.45–1.29 (m, 2H, CH_2), 0.94 (t, J = 7.4 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 144.7, 144.4, 142.3, 138.2, 138.0, 132.8,

© 2019 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2020, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201901284 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

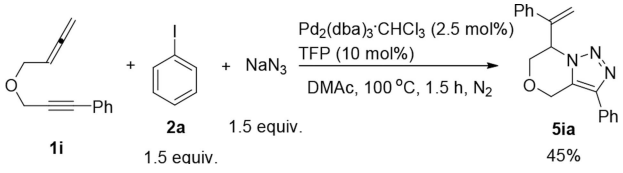
129.9, 129.0, 128.4, 127.7, 127.4, 125.6, 124.1, 116.8, 59.3, 46.7, 41.2, 30.9, 24.5, 22.2, 21.4, 21.3, 13.7; IR (neat) ν (cm^{-1}) 3041, 2957, 2929, 2860, 1597, 1579, 1493, 1457, 1353, 1306, 1293, 1266, 1221, 1169, 1091, 1055; MS (70 ev, EI) m/z (%) 451 ($M^+ + 1$, 4.15), 450 (M^+, 13.31), 295 (100); HRMS calcd for $C_{25}H_{30}N_4O_2S$ (M^+): 450.2089, found: 450.2089.

5-Benzyl-7-(1-(4-fluorophenyl)vinyl)-3-phenyl-4,5,6,7-tetrahydro-[1,2,3]triazolo[1,5-*a*]pyrazine 5hh



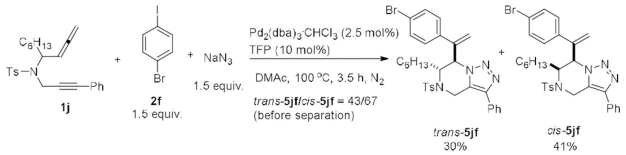
Following **Typical Procedure II**, the reaction of $Pd_2(dba)_3 \cdot CHCl_3$ (0.0053 g, 0.005 mmol), TFP (0.0044 g, 0.02 mmol), NaN_3 (0.0198 g, 0.3 mmol), **1h** (0.0550 g, 0.2 mmol)/DMAc (1 mL), and **2h** (0.0669 g, 0.3 mmol)/DMAc (1 mL) afforded **5hh** (0.0393 g, 48%) (eluent: petroleum ether/ethyl acetate/DCM = 6/1/1 (640 mL)): solid; m.p. 147.6–148.2 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.77–7.64 (m, 2H, ArH), 7.44 (t, J = 7.5 Hz, 2H, ArH), 7.38–7.20 (m, 6H, ArH), 7.19–7.08 (m, 2H, ArH), 6.94 (t, J = 8.7 Hz, 2H, ArH), 5.54 (t, J = 4.2 Hz, 1H, CH), 5.39 (s, one proton of =CH_2), 4.78 (s, one proton of =CH_2), 4.14 (d, J = 14.7 Hz, one proton of CH_2), 3.85 (d, J = 12.6 Hz, one proton of CH_2), 3.80 (d, J = 10.8 Hz, one proton of CH_2), 3.62 (d, J = 13.2 Hz, one proton of CH_2), 2.96–2.80 (m, 2H, CH_2); ^{13}C NMR (75 MHz, $CDCl_3$) δ 162.4 (d, J = 246.1 Hz), 144.9, 141.2, 136.8, 134.9 (d, J = 3.5 Hz), 131.1, 129.1, 128.80, 128.75, 128.73 (d, J = 9.7 Hz), 128.5, 127.63, 127.61, 126.2, 117.2, 115.3 (d, J = 20.7 Hz), 61.3, 60.3, 52.6, 49.6; ^{19}F NMR (282 MHz, $CDCl_3$) δ −114.5 (s, 1F); IR (neat) ν (cm^{-1}) 3062, 3028, 2811, 2757, 1602, 1509, 1450, 1364, 1302, 1234, 1162, 1098, 1070, 1029, 1008; MS (70 ev, EI) m/z (%) 411 ($M^+ + 1$, 0.55), 410 (M^+, 15.89), 91 (100); Anal. Calcd for $C_{26}H_{23}FN_4$: C 76.08, H 5.65, N 13.65. Found: C 75.76, H 5.69, N 13.60.

3-Phenyl-7-(1-phenylvinyl)-6,7-dihydro-4*H*-[1,2,3]triazolo- [5,1-*c*]-[1,4]oxazine 5ia



Following **Typical Procedure II**, the reaction of $Pd_2(dba)_3 \cdot CHCl_3$ (0.0127 g, 0.0125 mmol), TFP (0.0115 g, 0.05 mmol), NaN_3 (0.0490 g, 0.75 mmol), **1i**[25] (0.0920 g, 0.5 mmol)/DMAc (2 mL), and **2a** (0.1528 g, 0.75 mmol)/DMAc (3 mL) afforded **5ia** (0.0683 g, 45%). (The crude residue was first purified by column chromatography on silica gel (eluent: petroleum ether/ethyl acetate/DCM = 5/1/1 (560 mL)) to afford impure **5ia**, which was further purified by column chromatography on silica gel (eluent: DCM/diethyl ether = 150/1 (450 mL)) to afford pure **5ia**): solid; m.p. 173.8–174.7 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.75–7.61 (m, 2H, ArH), 7.51–7.41 (m, 2H, ArH), 7.40–7.28 (m, 6H, ArH), 5.59 (t, J = 3.6 Hz, 1H, CH), 5.49 (s, one proton of =CH_2), 5.17 (d, J = 15.3 Hz, one proton of OCH_2), 5.03 (d, J = 15.0 Hz, one proton of OCH_2), 4.71 (d, J = 0.9 Hz, one proton of =CH_2), 4.06–3.93 (m, 2H, OCH_2); ^{13}C NMR (75 MHz, $CDCl_3$) δ 144.5, 140.9, 138.3, 130.8, 128.9, 128.7, 128.3, 127.8, 127.6, 126.8, 126.0, 116.9, 66.7, 63.3, 59.2; IR (neat) ν (cm^{-1}) 3048, 2978, 2877, 2840, 1634, 1609, 1579, 1495, 1460, 1442, 1368, 1293, 1239, 1227, 1169, 1160, 1091, 1058, 1026, 1011; MS (70 ev, EI) m/z (%) 304 ($M^+ + 1$, 9.10), 303 (M^+, 41.25), 118 (100); Anal. Calcd for $C_{19}H_{17}N_3O$: C 75.23, H 5.65, N 13.85. Found: C 75.26, H 5.71, N 13.92.

Trans-7-(1-(4-bromophenyl)vinyl)-6-hexyl-3-phenyl-5-tosyl-4,5,6,7-tetrahydro-[1,2,3]triazolo[1,5-*a*] pyrazine (*trans*-5jf) and *cis*-7-(1-(4-bromophenyl) vinyl)-6-hexyl-3-phenyl-5-tosyl-4,5,6,7-tetrahydro-[1,2,3]triazolo[1,5-*a*]pyrazine (*cis*-5jf)



Following **Typical Procedure II**, the reaction of $Pd_2(dba)_3 \cdot CHCl_3$ (0.0132 g, 0.0125 mmol), TFP (0.0115 g, 0.05 mmol), NaN_3 (0.0490 g, 0.75 mmol), **1j** (0.2102 g, 0.5 mmol)/DMAc (2 mL), and **2f** (0.2119 g, 0.75 mmol)/DMAc (3 mL) afforded *trans*-**5jf** (0.0918 g, 30%) and *cis*-**5jf** (0.1275 g, 41%). The crude residue was first purified by column chromatography on silica gel (eluent: petroleum ether/ethyl acetate = 15/1 (450 mL) to 10/1 (770 mL)) to afford a part of pure *trans*-**5jf**, a part of pure *cis*-**5jf**, and a mixture of *trans*-**5jf** and *cis*-**5jf**, which was further purified by column chromatography on silica gel (petroleum ether/ethyl acetate = 10/1 (550 mL)) to afford the second part of pure *trans*-**5jf** and the second part of pure *cis*-**5jf**)).

trans-**5jf**: Less polar; oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.83–7.72 (m, 2H, ArH), 7.66 (d, J = 8.4 Hz, 2H, ArH), 7.62–7.45 (m, 4H, ArH), 7.45–7.27 (m, 5H, ArH), 5.73 (s, 1H, CH), 5.33 (s, one proton of =CH_2), 5.21 (d, J = 16.2 Hz, one proton of NCH_2), 4.40 (d, J = 16.2 Hz, one proton of NCH_2), 4.24 (d, J = 1.5 Hz, one proton of =CH_2), 4.21–4.12 (m, 1H, NCH), 2.42 (s, 3H, CH_3), 1.43–0.92 (m, 10H, $CH_2 \times 5$), 0.82 (t, J = 7.1 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 144.2, 144.1, 141.3, 136.8, 136.3, 131.9, 130.4, 129.8, 129.0, 128.6, 128.1, 127.1, 126.0, 124.9, 122.6, 116.4, 61.8, 54.9, 38.4, 31.3, 29.2, 28.5, 26.2, 22.3, 21.5, 13.9; IR (neat) ν (cm^{-1}) 2946, 2928, 2857, 1597, 1490, 1446, 1349, 1305, 1243, 1168, 1093, 1008; MS (70 ev, EI) m/z (%) 620 (M^+(^{81}Br), 4.58), 618 (M^+(^{79}Br), 4.01), 91 (100); HRMS calcd for $C_{32}H_{35}BrN_4O_2S$ (M^+(^{79}Br)): 618.1664, found: 618.1664.

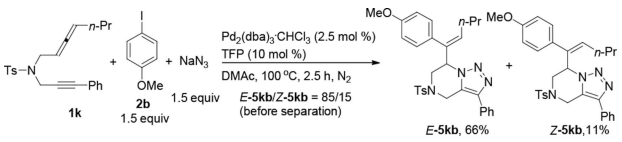
cis-**5jf**: More polar; solid; m.p. 133.9–134.8 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.75 (d, J = 7.5 Hz, 2H, ArH), 7.61–7.34 (m, 7H, ArH), 7.26–7.06 (m, 4H, ArH), 5.69

© 2019 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2020, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201901284 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

(s, 1H, CH), 5.37 (d, J = 17.4 Hz, one proton of NCH_2), 5.36 (s, one proton of $=CH_2$), 5.14 (d, J = 4.5 Hz, one proton of $=CH_2$), 4.54 (d, J = 18.0 Hz, one proton of NCH_2), 4.10 (dt, J_1 = 11.1 Hz, J_1 = 5.1 Hz, one proton of NCH), 2.37 (s, 3H, CH_3), 1.66–0.96 (m, 10H, $CH_2 \times 5$), 0.83 (t, J = 6.9 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 144.3, 141.6, 139.6, 137.2, 136.6, 132.0, 130.2, 129.9, 128.9, 128.0, 127.8, 126.5, 126.4, 126.2, 122.7, 118.8, 61.3, 52.7, 37.6, 31.3, 28.2, 25.1, 24.9, 22.2, 21.3, 13.8; IR (neat) ν (cm^{-1}) 3060, 2951, 2927, 2857, 1597, 1490, 1465, 1349, 1305, 1164, 1092, 1073, 1047, 1007; MS (70 ev, EI) m/z (%) 592 [(M^+ (^{81}Br)–N_2), 2.29], 590 ((M^+ (^{79}Br)–N_2), 2.41), 507 [(M^+ (^{81}Br)–$C_6H_{13}N_2$), 38.21], 505 [(M^+ (^{79}Br)–$C_6H_{13}N_2$), 35.81], 91 (100); Anal. Calcd for $C_{32}H_{35}BrN_4O_2S$: C 62.03, H 5.69, N 9.04. Found: C 61.98, H 5.64, N 9.02.

(*E*)-7-(1-(4-methoxyphenyl)pent-1-en-1-yl)-3-phenyl-5-tosyl-[1,2,3]triazolo[1,5-*a*]pyrazine (*E*-5 kb) and (*Z*)-7-(1-(4-methoxyphenyl)pent-1-en-1-yl)-3-phenyl-5-tosyl- [1,2,3]triazolo [1,5-*a*] pyrazine (*Z*-5 kb)

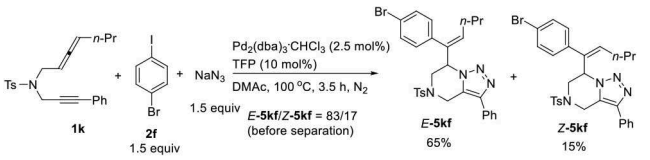


Following **Typical Procedure II**, the reaction of $Pd_2(dba)_3 \cdot CHCl_3$ (0.0082 g, 0.008 mmol), TFP (0.0073 g, 0.032 mmol), NaN_3 (0.0310 g, 0.48 mmol), **1 k** (0.1224 g, 0.32 mmol)/DMAc (1 mL), and **2 b** (0.1125 g, 0.48 mmol)/DMAc (2 mL) afforded ***E*-5 kb** (0.1126 g, 66%) and impure ***Z*-5 kb** (0.0253 g) (eluent: petroleum ether/ethyl acetate/DCM = 4.5/1/1 (650 mL)). Impure ***Z*-5 kb** was further purified by column chromatography on silica gel (petroleum ether/ethyl acetate/DCM = 8/1/1 (400 mL)) to afford pure ***Z*-5 kb** (0.0194 g, 11%).

***Z*-5 kb**: Less polar; oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.70–7.61 (m, 4H, ArH), 7.53–7.43 (m, 2H, ArH), 7.42–7.27 (m, 3H, ArH), 6.78 (d, J = 9.0 Hz, 2H, ArH), 6.68 (d, J = 8.7 Hz, 2H, ArH), 5.89 (t, J = 7.5 Hz, 1H, CH), 5.55 (dd, J_1 = 9.5 Hz, J_2 = 4.7 Hz, one proton of=CH), 4.94 (dd, J_1 = 15.6 Hz, J_2 = 1.2 Hz, one proton of NCH_2), 4.20 (d, J = 15.6 Hz, one proton of NCH_2), 3.95 (ddd, J_1 = 13.4 Hz, J_2 = 4.7 Hz, J_3 = 1.1 Hz, one proton of NCH_2), 3.72 (s, 3H, CH_3), 3.19 (dd, J_1 = 13.2 Hz, J_2 = 9.6 Hz, one proton of NCH_2), 2.42 (s, 3H, CH_3), 2.33–2.00 (m, 2H, CH_2), 1.65–1.42 (m, 2H, CH_2), 0.98 (t, J = 7.4 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 158.9, 144.7, 142.2, 137.9, 133.8, 133.6, 132.0, 130.5, 130.1, 129.4, 129.0, 128.1, 127.3, 126.2, 125.8, 113.6, 56.2, 55.1, 46.9, 42.5, 30.2, 22.6, 21.5, 13.9; IR (neat) ν (cm^{-1}) 2959, 2931, 2873, 1608, 1510, 1456, 1352, 1287, 1246, 1169, 1091, 1032, 1008; MS (70 ev, EI) m/z (%) 529 (M^+ + 1, 4.43), 528 (M^+, 16.57), 373 (100); HRMS calcd for $C_{30}H_{32}N_4O_3S$ (M^+): 528.2195, found: 528.2194.

***E*-5 kb**: More polar; oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.75–7.58 (m, 4H, ArH), 7.44 (t, J = 7.5 Hz, 2H, ArH), 7.39–7.30 (m, 3H, ArH), 7.11 (d, J = 8.4 Hz, 2H, ArH), 6.88 (d, J = 8.7 Hz, 2H, ArH), 5.34 (t, J = 4.2 Hz, 1H, CH), 5.17 (t, J = 7.2 Hz, 1H, =CH), 4.67 (d, J = 15.3 Hz, one proton of NCH_2), 4.31 (d, J = 15.0 Hz, one proton of NCH_2), 3.79 (s, 3H, CH_3), 3.64 (dd, J_1 = 12.6 Hz, J_2 = 4.8 Hz, one proton of NCH_2), 3.32 (dd, J_1 = 12.8 Hz, J_2 = 4.4 Hz, one proton of NCH_2), 2.42 (s, 3H, CH_3), 1.96–1.78 (m, 2H, CH_2), 1.41–1.21 (m, 2H, CH_2), 0.80 (t, J = 7.4 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 158.9, 144.5, 141.5, 135.3, 133.7, 132.9, 130.6, 130.4, 130.0, 128.8, 128.6, 127.9, 127.4, 126.0, 125.7, 113.8, 62.0, 55.1, 46.2, 42.4, 30.6, 22.3, 21.4, 13.6; IR (neat) ν (cm^{-1}) 3055, 2958, 2931, 2870, 2834, 1608, 1511, 1456, 1353, 1289, 1247, 1172, 1154, 1092, 1033, 1008; MS (70 ev, EI) m/z (%) 529 (M^+ + 1, 3.79), 528 (M^+, 10.47), 373 (100); HRMS calcd for $C_{30}H_{32}N_4O_3S$ (M^+): 528.2195, found: 528.2192.

(*E*)-7-(1-(4-bromophenyl)pent-1-en-1-yl)-3-phenyl-5-tosyl-4,5,6,7-tetrahydro-[1,2,3]triazolo[1,5-a] pyrazine (*E*-5 kf) and (*Z*)-7-(1-(4-bromophenyl) pent-1-en-1-yl)-3-phenyl-5-tosyl-4,5,6,7-tetrahydro-[1,2,3]triazolo[1,5-a]pyrazine (*Z*-5 kf)



Following **Typical Procedure II**, the reaction of $Pd_2(dba)_3 \cdot CHCl_3$ (0.0126 g, 0.0125 mmol), TFP (0.0115 g, 0.05 mmol), NaN_3 (0.0490 g, 0.75 mmol), **1 k** (0.1901 g, 0.5 mmol)/DMAc (2 mL), and **2 f** (0.2125 g, 0.75 mmol)/DMAc (3 mL) afforded ***E*-5 kf** (0.1894 g, 65%) and ***Z*-5 kf** (0.0437 g, 15%) (eluent: petroleum ether/ethyl acetate/DCM = 9/1/1 (330 mL) to 8/1/1 (550 mL)).

***Z*-5 kf**: Less polar. solid; m.p. 189.5–190.2 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.68–7.60 (m, 4H, ArH), 7.47 (t, J = 7.5 Hz, 2H, ArH), 7.42–7.25 (m, 5H, ArH), 6.74 (d, J = 8.4 Hz, 2H, ArH), 5.92 (t, J = 7.5 Hz, 1H, CH), 5.59 (dd, J_1 = 8.8 Hz, J_2 = 5.0 Hz, one proton of $=CH_2$), 4.84 (d, J = 15.3 Hz, one proton of NCH_2), 4.21 (d, J = 15.6 Hz, one proton of NCH_2), 3.90 (dd, J_1 = 13.1 Hz, J_2 = 4.4 Hz, one proton of NCH_2), 3.19 (dd, J_1 = 13.2 Hz, J_2 = 9.3 Hz, one proton of NCH_2), 2.42 (s, 3H, CH_3), 2.33–2.06 (m, 2H, CH_2), 1.67–1.46 (m, 2H, CH_2), 0.99 (t, J = 7.4 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 144.8, 142.3, 139.0, 138.5, 133.4, 133.3, 131.4, 130.3, 130.2, 130.0, 129.0, 128.2, 127.3, 126.2, 125.8, 121.6, 56.0, 47.0, 42.5, 30.2, 22.5, 21.5, 13.9; IR (neat) ν (cm^{-1}) 3061, 2958, 2870, 1597, 1486, 1454, 1447, 1352, 1306, 1167, 1091, 1074, 1008; MS (70 ev, EI) m/z (%) 578 (M^+ (^{81}Br), 16.72), 576 (M^+(^{79}Br), 16.82), 91 (100); Anal. Calcd for $C_{29}H_{29}BrN_4O_2S$: C 60.31, H 5.06, N 9.70. Found: C 59.66, H 5.07, N 9.61. HRMS calcd for $C_{29}H_{29}N_4O_2SBr$ (M^+ (^{79}Br)): 576.1195, found: 576.1196.

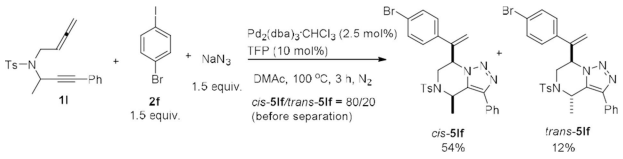
***E*-5 kf**: More polar. solid; m.p. 206.0–206.5 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.78–7.58 (m, 4H, ArH), 7.57–7.42 (m, 4H, ArH), 7.42–7.30 (m, 3H, ArH), 7.08 (d, J = 8.4 Hz, 2H, ArH), 5.36 (t, J = 3.6 Hz, 1H, CH), 5.25 (t, J = 7.4 Hz, 1H, CH), 4.69 (d, J = 15.0 Hz, one proton of NCH_2), 4.30 (d, J = 15.0 Hz, one proton of CH_2), 3.62 (dd, J_1 = 12.6 Hz, J_2 = 4.8 Hz, one proton of CH_2), 3.31 (dd, J_1 = 12.8 Hz, J_2 = 4.2 Hz, one proton of CH_2), 2.43 (s, 3H, CH_3), 1.98–1.77 (m,

© 2019 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2020, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201901284 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

2H, CH_2), 1.43–1.19 (m, 2H, CH_2), 0.81 (t, J=7.4 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 144.6, 141.5,135.4, 134.7, 134.3, 132.7, 131.6, 131.2, 130.2, 130.1, 128.9, 128.0, 127.4, 126.0, 125.7, 121.8, 61.6, 46.2, 42.4, 30.6, 22.2, 21.4, 13.5; IR (neat) ν (cm^{-1}) 3067, 2959, 2929, 2870, 1597, 1486, 1456, 1444, 1352, 1305, 1265, 1227, 1171, 1156, 1092, 1070, 1009; MS (70 ev, EI) m/z (%) 578 (M^+(^{81}Br), 18.70), 576 (M^+(^{79}Br), 17.79), 91 (100). Anal. Calcd for $C_{29}H_{29}N_4O_2SBr$: C 60.31, H 5.06, N 9.70. Found: C 59.73, H 5.07, N 9.62. HRMS calcd for $C_{29}H_{29}N_4O_2SBr$ (M^+ (^{79}Br)): 576.1195, found: 576.1192.

Cis-7-(1-(4-bromophenyl)vinyl)-4-methyl-3-phenyl-5-tosyl-4,5,6,7-tetrahydro-[1,2,3]triazolo[1,5-*a*] pyrazine (*cis*-5 lf) and *trans*-7-(1-(4-bromophenyl) vinyl)-4-methyl-3-phenyl-5-tosyl-4,5,6,7-tetrahydro-[1,2,3]triazolo[1,5-*a*]pyrazine (*trans*-5 lf)

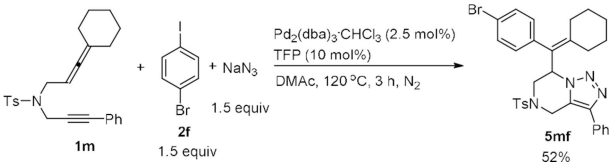


Following **Typical Procedure II**, the reaction of $Pd_2(dba)_3$·$CHCl_3$ (0.0131 g, 0.0125 mmol), TFP (0.0116 g, 0.05 mmol), NaN_3 (0.0487 g, 0.75 mmol), **1 l** (0.1757 g, 0.5 mmol)/DMAc (2 mL), and **2 f** (0.2121 g, 0.75 mmol)/DMAc (3 mL) afforded *cis*-**5 lf** (0.1472 g, 54%) and *trans*-**5 lf** (0.0328 g, 12%)(eluent: petroleum ether/ethyl acetate/DCM = 5/1/1 (700 mL)).

cis-**5 lf**: Less polar; solid; m.p. 182.7–183.7 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.78 (d, J=7.2 Hz, 2H, ArH), 7.61 (d, J=8.4 Hz, 2H, ArH), 7.50 (t, J=7.4 Hz, 2H, ArH), 7.46–7.35 (m, 3H, ArH), 7.22 (d, J=7.8 Hz, 2H, ArH), 6.93 (d, J=8.4 Hz, 2H, ArH), 5.74 (q, J=6.8 Hz, 1H, CH), 5.62 (s, one proton of CH_2=), 5.37 (s, one proton of CH_2=), 5.00 (dd, J_1=10.7 Hz, J_2=5.6 Hz, 1H, CH), 4.17 (dd, J_1=15.0 Hz, J_2=5.4 Hz, one proton of NCH_2), 3.43 (dd, J_1=15.0 Hz, J_2=10.8 Hz, one proton of NCH_2), 2.39 (s, 3H, CH_3), 1.37 (d, J=6.9 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 144.6, 142.5, 141.4, 136.5, 136.2, 131.8, 131.1, 130.3, 130.1, 129.0, 128.7, 128.2, 126.6, 126.3, 122.7, 120.7, 59.3, 47.7, 42.6, 21.5, 19.5; IR (KBr) ν (cm^{-1}) 3054, 2983, 2929, 2860, 1597, 1489, 1446, 1375, 1349, 1328, 1305, 1167, 1156, 1120, 1091, 1073, 1049, 1006; MS (70 ev, EI) m/z (%) 550 (M^+ (^{81}Br), 8.30), 548 (M^+(^{79}Br), 7.48), 56 (100); HRMS calcd for $C_{27}H_{25}BrN_4O_2S$ (M^+(^{79}Br)): 548.0882, found: 548.0880.

trans-**5 lf**: More polar; oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.80 (d, J=7.2 Hz, 2H, ArH), 7.64 (d, J=8.1 Hz, 2H, ArH), 7.57–7.45 (m, 4H, ArH), 7.43–7.34 (m, 1H, ArH), 7.33–7.18 (m, 4H, ArH), 5.85 (q, J=6.7 Hz, 1H, CH), 5.74 (d, J=3.0 Hz, 1H, CH_2=), 5.32 (s, 1H, CH_2=), 4.24 (s, 1H, CH), 3.94 (d, J=13.8 Hz, one proton of NCH_2), 3.60 (dd, J_1=14.0 Hz, J_2=4.1 Hz, one proton of NCH_2), 2.43 (s, 3H, CH_3), 1.17 (d, J=6.6 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 144.1, 143.3, 141.1, 137.2, 136.7, 131.8, 131.1, 130.4, 129.8, 129.0, 128.6, 128.2, 127.2, 126.5, 122.5, 116.8, 58.5, 48.1, 40.7, 21.5, 17.0; IR (neat) ν (cm^{-1}) 3055, 2983, 2920, 2856, 1597, 1491, 1446, 1349, 1305, 1265, 1170, 1155, 1091, 1056, 1008; MS (70 ev, EI) m/z (%) 550 (M^+(^{81}Br), 9.46), 548 (M^+(^{79}Br), 9.09), 56 (100); HRMS calcd for $C_{27}H_{25}BrN_4O_2S$ (M^+(^{79}Br)): 548.0882, found: 548.0883.

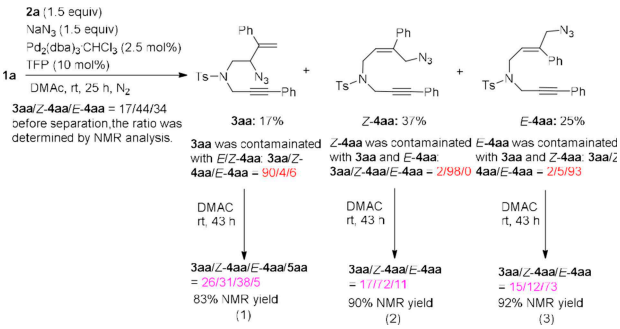
7-((4-Bromophenyl)(cyclohexylidene)methyl)-3-phenyl-5-tosyl-4,5,6,7-tetrahydro-[1,2,3]triazolo [1,5-*a*]pyrazine 5 mf



Following **Typical Procedure II**, the reaction of $Pd_2(dba)_3$·$CHCl_3$ (0.0078 g, 0.0075 mmol), TFP (0.0072 g, 0.03 mmol), NaN_3 (0.0294 g, 0.45 mmol), **1 m** (0.1233 g, 0.3 mmol)/DMAc (1 mL), and **2 f** (0.1275 g, 0.45 mmol)/DMAc (2 mL) afforded **5 mf** (0.0960 g, 52%) (eluent: petroleum ether/ethyl acetate/DCM = 7/1/1 (450 mL)): solid; m.p. 180.9–181.9 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, d_6-DMSO) δ 7.81 (d, J=8.1 Hz, 2H, ArH), 7.63 (d, J=7.5 Hz, 2H, ArH), 7.56–7.12 (m, 8H, ArH), 6.81–6.22 (m, 1H, ArH), 5.83 (t, J=5.1 Hz, 1H, NCH), 4.52 (d, J=15.6 Hz, one proton of NCH_2), 4.39 (d, J=15.6 Hz, one proton of NCH_2), 3.81 (dd, J_1=12.9 Hz, J_2=3.9 Hz, one proton of NCH_2), 3.42–3.37 (m, one proton of CH_2), 2.51–2.33 (m, 5H, CH_3+CH_2), 1.89–1.38 (m, 8H, CH_2×3+CH_2);^{13}C NMR (75 MHz, d_6-DMSO) δ145.5, 145.3, 141.8, 138.0, 133.9, 131.7, 131.2, 131.1, 130.0, 128.8, 128.4, 127.4, 126.9, 126.8, 121.2, 56.7, 48.5, 43.0, 33.5, 31.0, 28.82, 28.77, 27.0, 22.0; IR (neat) ν (cm^{-1}) 2928, 2853, 1596, 1581, 1486, 1447, 1352, 1170, 1091, 1070, 1008; MS (70 ev, EI) m/z (%) 604 (M(^{81}Br)$^+$, 4.57), 602 (M (^{79}Br)$^+$, 4.68), 91 (100); Anal. Calcd for $C_{31}H_{31}BrN_4O_2S$: C 61.69, H 5.18, N 9.28. Found: C 61.51, H 5.19, N 9.33.

Control Experiments

Isolation and Observation of the Interconversion of the Three Allylic Azide Precursors 3 aa, Z-4 aa, and E-4 aa



To a flame-dried Schlenk tube were added $Pd_2(dba)_3$·$CHCl_3$ (0.0257 g, 0.025 mmol), TFP (0.0232 g, 0.1 mmol), NaN_3 (0.0979 g, 1.5 mmol), **1 a** (0.3378 g, 1.0 mmol)/DMAc (4 mL), and **2 a** (0.3067 g, 1.5 mmol)/DMAc (6 mL) sequentially under N_2 atmosphere. The reaction mixture was stirred at room temperature for 25 h until **1 a** was completely consumed as

© 2019 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2020, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201901284 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

monitored by TLC. Then the reaction was quenched with H_2O (15 mL), and extracted with ethyl acetate (15 mL × 3). The combined organic layer was washed with H_2O (15 mL) and brine (15 mL), and dried over anhydrous Na_2SO_4. After filtration and evaporation of the solvent, the crude residue was purified by chromatography on silica gel (eluent: petroleum ether/ethyl acetate = 20/1 (1000 mL)) to afford **3 aa** (0.0790 g, 17%, **3 aa** was contamainated with *Z*/*E*-**4 aa**: **3 aa**/*Z*-**4 aa**/*E*-**4 aa** = 90/4/6), *Z*-**4 aa** (0.1680 g, 37%, *Z*-**4 aa** was contamainated with **3 aa**/*E*-**4 aa**: **3 aa**/*Z*-**4 aa**/*E*-**4 aa** = 2/98/0), and *E*-**4 aa** (0.1141 g, 25%, *E*-**4 aa** was contamainated with **3 aa**/*Z*-**4 aa**:**3 aa**/*Z*-**4 aa**/*E*-**4 aa** = 2/5/93):

3 aa (**3 aa**/*Z*-**4 aa**/*E*-**4 aa** = 90/4/6): oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.66 (d, *J* = 8.1 Hz, 2H, ArH), 7.52–7.43 (m, 2H, ArH), 7.40–7.13 (m, 8H, ArH), 7.06–6.95 (m, 2H, ArH), 5.58 (s, one proton of =CH_2), 5.47 (s, one proton of =CH_2), 4.92 (dd, J_1 = 9.3 Hz, J_2 = 3.9 Hz, 1H, CH), 4.49 (d, *J* = 18.6 Hz, one proton of NCH_2), 4.41 (d, *J* = 18.6 Hz, one proton of NCH_2), 3.38 (dd, J_1 = 14.9 Hz, J_2 = 4.1 Hz, one proton of NCH_2), 3.21 (dd, J_1 = 14.9 Hz, J_2 = 9.5 Hz, one proton of NCH_2), 2.29 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 144.0, 143.7, 138.2, 135.2, 131.4, 129.4, 128.6, 128.4, 128.2, 128.0, 127.7, 126.5, 121.8, 116.5, 85.6, 81.8, 65.4, 50.1, 39.5, 21.3; IR (neat) ν (cm^{-1}) 3057, 2922, 2856, 2105, 1598, 1570, 1490, 1443, 1349, 1258, 1161, 1092, 1028; MS (ESI, m/z) 479 ($M+Na^+$); HRMS calcd. for $C_{26}H_{24}N_4O_2SNa$ ($M+Na^+$): 479.1518; Found: 479.1520. The following signals are discernible for *Z*-**4 aa**: ^{1}H NMR (300 MHz, $CDCl_3$) δ 6.07 (t, *J* = 7.4 Hz, 1H, =CH), 2.34 (s, 3H, CH_3); The following signals are discernible for *E*-**4 aa**: ^{1}H NMR (300 MHz, $CDCl_3$) δ 5.78 (t, *J* = 6.9 Hz, 1H, =CH), 2.24 (s, 3H, CH_3).

***Z*-4 aa** (**3 aa**/*Z*-**4 aa**/*E*-**4 aa** = 2/98/0): oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.80 (d, *J* = 8.1 Hz, 2H, ArH), 7.43–7.20 (m, 10H, ArH), 7.13–7.00 (m, 2H, ArH), 6.07 (t, *J* = 7.4 Hz, 1H, =CH), 4.36 (s, 2H, NCH_2), 4.30 (s, 2H, NCH_2), 4.16 (d, *J* = 7.2 Hz, 2H, NCH_2), 2.35 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 143.8, 139.3, 139.1, 135.6, 131.4, 129.6, 128.6, 128.5, 128.3, 128.2, 127.8, 126.9, 126.2, 121.8, 86.1, 81.5, 48.4, 44.1, 37.1, 21.4; IR (neat) ν (cm^{-1}) 3057, 3028, 2922, 2096, 1598, 1490, 1443, 1348, 1246, 1162, 1091; MS (ESI, m/z) 479 ($M+Na^+$); HRMS calcd. for $C_{26}H_{24}N_4O_2SNa$ ($M+Na^+$): 479.1518; Found: 479.1526. The following signals are discernible for **3 aa**: ^{1}H NMR (300 MHz, $CDCl_3$) δ 5.59 (s, one proton of =CH_2), 5.47 (s, one proton of =CH_2), 2.30 (s, 3H, CH_3).

E-**4 aa** (**3 aa**/*Z*-**4 aa**/*E*-**4 aa** = 2/5/93): oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.67 (d, *J* = 8.1 Hz, 2H, ArH), 7.29–7.12 (m, 10H, ArH), 6.89–6.79 (m, 2H, ArH), 5.79 (t, *J* = 6.9 Hz, 1H, =CH), 4.28 (s, 2H, NCH_2), 4.01 (s, 2H, NCH_2), 3.92 (d, *J* = 6.6 Hz, 2H, NCH_2), 2.25 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 143.5, 140.3, 136.4, 135.6, 131.3, 129.4, 128.4, 128.2, 128.1, 127.93, 127.86, 127.6, 125.6, 121.9, 85.6, 81.4, 57.6, 44.7, 37.0, 21.3 ; IR (neat) ν (cm^{-1}) 3056, 3024, 2922, 2101, 1598, 1490, 1443, 1348, 1258, 1163, 1091; MS (ESI, m/z) 479 ($M+Na^+$); HRMS calcd. for $C_{26}H_{24}N_4O_2SNa$ ($M+Na^+$): 479.1518; Found: 479.1524. The following signals are discernible for **3 aa**: ^{1}H NMR (300 MHz, $CDCl_3$) δ 5.59 (s, one proton of =CH_2), 5.47 (s, one proton of =CH_2), 2.30 (s, 3H, CH_3); The following signals are discernible for *Z*-**4 aa**: ^{1}H NMR (300 MHz, $CDCl_3$) δ 6.07 (t, *J* = 7.7 Hz, 1H, =CH), 2.35 (s, 3H, CH_3).

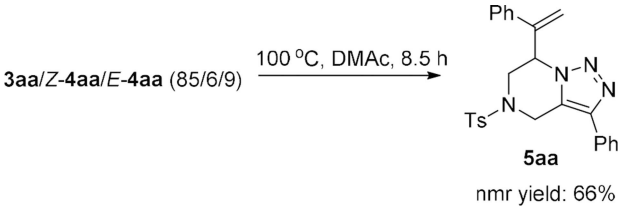
(1) To a vial were added the mixture of **3 aa**/*Z*-**4 aa**/*E*-**4 aa** (90/4/6, 0.0790 g, 0.17 mmol) and DMAc (2 mL). After being stirred at room temperature for 43 h, the reaction mixture was quenched with H_2O (5 mL), and extracted with ethyl acetate (5 mL × 3). The combined organic layer was washed with H_2O (5 mL) and brine (5 mL), and dried over anhydrous Na_2SO_4. After filtration and evaporation of the solvent, 83% NMR yield of the mixture of **3 aa**/*Z*-**4 aa**/*E*-**4 aa**/**5 aa** (26/31/38/5) was determined by ^{1}H NMR analysis of the crude residue using mesitylene (16 μL) as the internal standard.

(2) To a vial were added the mixture of **3 aa**/*Z*-**4 aa**/*E*-**4 aa** (2/98/0, 0.1680 g, 0.37 mmol) and DMAC (2 mL). After being stirred at room temperature for 43 h, the reaction mixture was quenched with H_2O (5 mL), and extracted with ethyl acetate (5 mL × 3). The combined organic layer was washed with H_2O (5 mL) and brine (5 mL), and dried over anhydrous Na_2SO_4. After filtration and evaporation of the solvent, 90% NMR yield of the mixture of **3 aa**/*Z*-**4 aa**/*E*-**4 aa** (17/72/11) was determined by ^{1}H NMR analysis of the crude residue using mesitylene (17 μL) as the internal standard.

(3) To a vial were added the mixture of **3 aa**/*Z*-**4 aa**/*E*-**4 aa** (2/5/93, 0.1141 g, 0.25 mmol) and DMAc (2 mL). After being stirred at room temperature for 43 h, the reaction mixture was quenched with H_2O (5 mL), and extracted with ethyl acetate (5 mL × 3). The combined organic layer was washed with H_2O (5 mL) and brine (5 mL), and dried over anhydrous Na_2SO_4. After filtration and evaporation of the solvent, 92% NMR yield of the mixture of **3 aa**/*Z*-**4 aa**/*E*-**4 aa** (15/12/73) was determined by ^{1}H NMR analysis of the crude residue using mesitylene (11.5 μL) as the internal standard.

The Transformation of the three Allylic Azide Precursors 3 aa and Z/E-4 aa with Different Ratios into 5 aa Exclusively without or with Pd-Catalyst

The transformation of the mixture of **3 aa**/Z-**4 aa**/E-**4 aa** (85/6/9) to **5 aa** without Pd-catlyst:



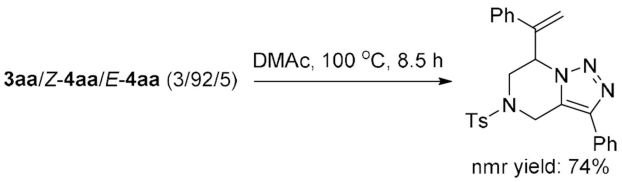
nmr yield: 66%

To a vial were added the mixture of **3 aa**/*Z*-**4 aa**/*E*-**4 aa** (85/6/9, 0.0557 g, 0.12 mmol) and DMAc (1.2 mL). The reaction mixture was stirred at 100 °C for 8.5 h until **3 aa**/*Z*-**4 aa**/*E*-**4 aa** were completely consumed as monitored by TLC. Then it was quenched with H_2O (5 mL), and extracted with ethyl acetate (5 mL × 3). The combined organic layer was washed with H_2O (5 mL) and brine (5 mL), and dried over anhydrous Na_2SO_4. After filtration and evaporation of the solvent, 66% NMR yield of **5 aa** was determined by ^{1}H NMR analysis of the crude residue using mesitylene (11.2 μL) as the internal standard.

The transformation of a moxture of **3 aa**/*Z*-**4 aa**/*E*-**4 aa** (3/92/5) to **5 aa** without Pd-catlyst:

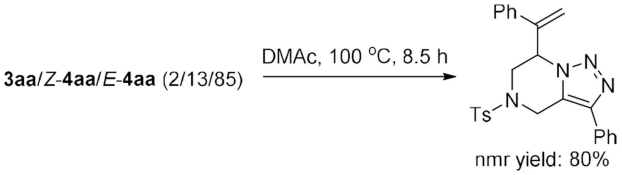
© 2019 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2020, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201901284 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License



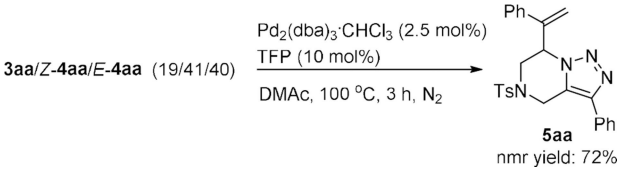
To a vial were added the mixture of **3aa**/*Z*-**4aa**/*E*-**4aa** (3/92/5, 0.1079 g, 0.24 mmol) and DMAc (2.4 mL). The reaction mixture was stirred at 100 °C for 8.5 h until **3aa**/*Z*-**4aa**/*E*-**4aa** were completely consumed as monitored by TLC. Then it was quenched with H_2O (5 mL), and extracted with ethyl acetate (5 mL × 3). The combined organic layer was washed with H_2O (5 mL) and brine (5 mL), and dried over anhydrous Na_2SO_4. After filtration and evaporation of the solvent, 74% NMR yield of **5aa** was determined by ^1H NMR analysis of the crude residue using mesitylene (10.9 µL) as the internal standard.

The transformation of a mixture of **3aa**/*Z*-**4aa**/*E*-**4aa** (2/13/85) to **5aa** without Pd-catlyst:



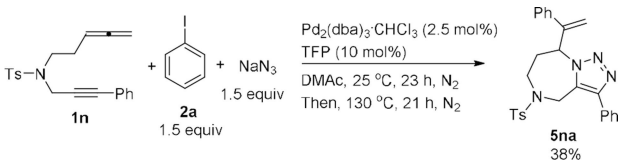
To a vial were added **3aa**/*Z*-**4aa**/*E*-**4aa** (2/13/85, 0.0699 g, 0.15 mmol) and DMAc (1.5 mL). The reaction mixture was stirred at 100 °C for 8.5 h until **3aa**/*Z*-**4aa**/*E*-**4aa** were completely consumed as monitored by TLC. Then it was quenched with H_2O (5 mL), and extracted with ethyl acetate (5 mL × 3). The combined organic layer was washed with H_2O (5 mL) and brine (5 mL), and dried over anhydrous Na_2SO_4. After filtration and evaporation of the solvent, 80% NMR yield of **5aa** was determined by ^1H NMR analysis of the crude residue using mesitylene (14 µL) as the internal standard.

The transformation of a mixture of **3aa**/*Z*-**4aa**/*E*-**4aa** (19/41/40) to **5aa** with Pd-catlyst:



To a flame-dried Schlenk tube were added $Pd_2(dba)_3 \cdot CHCl_3$ (0.0110 g, 0.0105 mmol), TFP (0.0099 g, 0.042 mmol), a mixture of **3aa**/*Z*-**4aa**/*E*-**4aa** (19/41/40, 0.1911 g, 0.42 mmol)/DMAc (4 mL) sequentially under N_2 atmosphere. The reaction mixture was stirred at 100 °C for 3 h until **3aa**/*Z*-**4aa**/*E*-**4aa** were completely consumed as monitored by TLC. Then it was quenched with H_2O (10 mL), and extracted with ethyl acetate (15 mL × 3). The combined organic layer was washed with H_2O (10 mL) and brine (10 mL), and dried over anhydrous Na_2SO_4. After filtration and evaporation of the solvent, 72% NMR yield of **5aa** was determined by ^1H NMR analysis of the crude residue using mesitylene (23 µL) as the internal standard.

3-Phenyl-8-(1-phenylvinyl)-5-tosyl-5,6,7,8-tetrahydro-4*H*-[1,2,3]-triazolo[1,5-*a*][1,4]diazepine 5na



To a flame-dried Schlenk tube were added $Pd_2(dba)_3 \cdot CHCl_3$ (0.0148 g, 0.0145 mmol), TFP (0.0137 g, 0.058 mmol), NaN_3 (0.0561 g, 0.86 mmol), **1n**[25] (0.2023 g, 0.58 mmol)/DMAc (2.8 mL), and **2a** (0.1755 g, 0.86 mmol)/DMAc (3 mL) sequentially under N_2 atmosphere. The reaction mixture was stirred at 25 °C for 23 h until **1n** was completely consumed as monitored by TLC. After that, reaction mixture was stirred at 130 °C for 21 h. The resulting mixture was then stop stirring, quenched with H_2O (10 mL), and extracted with ethyl acetate (15 mL × 3). The combined organic layer was washed with H_2O (10 mL) and brine (10 mL), and dried over anhydrous Na_2SO_4. After filtration and evaporation of the solvent, the crude residue was purified by column chromatography on silica gel (eluent: petroleum ether/ethyl acetate = 2/1 (800 mL)) to afford **5na** (0.1022 g, 38%): oil; ^1H NMR (300 MHz, $CDCl_3$) δ 7.86–7.78 (m, 2H, ArH), 7.59–7.43 (m, 5H, ArH), 7.43–7.30 (m, 5H, ArH), 7.19 (d, J = 8.1 Hz, 2H, ArH), 6.26 (s, one proton of =CH_2), 5.35 (d, J = 1.5 Hz, one proton of =CH_2), 5.29 (d, J = 16.5 Hz, one proton of CH_2), 4.37 (d, J = 1.8 Hz, one proton of CH_2), 4.16 (d, J = 16.2 Hz, one proton of CH_2), 3.87–3.75 (m, one proton of CH_2), 3.22–3.07 (m, one proton of CH_2), 2.38 (s, 3H, CH_3), 2.20–2.08 (m, one proton of CH_2), 2.07–1.93 (m, one proton of CH_2); ^{13}C NMR (75 MHz, $CDCl_3$) δ 145.6, 144.9, 143.9, 138.2, 135.7, 130.7, 130.4, 129.8, 129.0, 128.9, 128.6, 128.4, 127.9, 127.0, 126.7, 116.2, 63.3, 45.5, 41.8, 28.9, 21.5; IR (neat) ν (cm^{-1}) 3054, 3024, 2920, 2852, 1597, 1494, 1445, 1345, 1248, 1160, 1091; MS (70 ev, EI) m/z (%) 471 ($M^+ + 1$, 31.40), 470 (M^+, 100); HRMS calcd for $C_{27}H_{26}N_4O_2S$ (M^+): 470.1776, found: 470.1777.

Acknowledgements

Financial support from National Natural Science Foundation of China (21690063) is greatly appreciated. Shengming Ma is a Qiu Shi Adjunct Professor at Zhejiang University. We thank Mr. Guolin Wu in this group for reproducing the results of ***5af****,* ***5al*** *and* ***5da****.*

References

[1] a) A. Martínez, H. Gutiérrez-de-Terán, J. Brea, E. Raviña, M. Loza, M. Cadavid, F. Sanz, B. Vidal, V. Segarrad, E. Soteloa, *Bioorg. Med. Chem.* **2008**, *16*, 2103–2113; b) S. G. Agalave, S. R. Maujan, V. S. Pore, *Chem. Asian J.* **2011**, *6*, 2696–2718; c) R. S. Keri, S. A. Patil, S. Budagumpi, B. M. Nagaraja, *Chem. Biol. Drug Des.* **2015**, *86*, 410–423; d) H. Y. Hsieh, W. C. Lee, G. C. Senadi, W. P. Hu, J. J. Liang, T. R. Tsai, Y. W.

© 2019 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2020, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201901284 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Chou, K. K. Kuo, C. Y. Chen, J. J. Wang, *J. Med. Chem.* **2013**, *56*, 5422–5435.

[2] For selected reviews, see: a) K. Majumdar, K. Ray, *Synthesis*, **2011**, 3767–3783; b) R. S. Menon, V. Nair, In *Comprehensive Organic Synthesis*, 2nd edn. B. V. Elsevier, **2014**, *4*, 1281–1341; c) J. R. Johansson, T. Beke-Somfai, S. A. Stalsmeden, N. Kann, *Chem. Rev.* **2016**, *116*, 14726–14768; d) N. V. Sokolova, V. G. Nenajdenko, *RSC Adv.* **2013**, *3*, 16212–16242; e) A. Mohan, S. Sankararaman, *Isr. J. Chem.* **2012**, *52*, 92–104; f) T. M. A. Barlow, D. Tourwé, S. Ballet, *Eur. J. Org. Chem.* **2017**, 4678–4694.

[3] For selected reports on Cu-catalyzed intramolecular azide-alkyne click reaction, see: a) R. Huisgen, *Angew. Chem. Int. Ed.* **1963**, *2*, 633–645; b) V. Rostovtsev, L. Green, V. Fokin, K. Sharpless, *Angew. Chem. Int. Ed.* **2002**, *41*, 2596–2599; c) S. Chandrasekhar, M. Seenaiah, A. Kumar, C. R. Reddy, S. K. Mamidyala, C. G. Kumar, S. Balasubramanian, *Tetrahedron Lett.* **2011**, *52*, 806–808; d) A. S. Jadhav, Y. A. Pankhade, R. V. Anand, *J. Org. Chem.* **2018**, *83*, 8596–8606; e) R. H. Vekariya, R. Liu, J. Aubé, *Org. Lett.* **2014**, *16*, 1844–1847; f) C. Gagan, J. Keith, *Org. Lett.* **2011**, *13*, 2754–2757.

[4] For selected reports on Ru-catalyzed intramolecular azide-alkyne click reaction, see: a) L. Zhang, X. Chen, P. Xue, H. H. Y. Sun, I. D. Williams, K. B. Sharpless, V. V. Fokin, G. Jia, *J. Am. Chem. Soc.* **2005**, *127*, 15998–15999; b) L. A. Marcaurelle, E. Comer, S. Dandapani, J. R. Duvall, B. Gerard, S. Kesavan, M. D. Lee, H. Liu, J. T. Lowe, J. C. Marie, C. A. Mulrooney, B. A. Pandya, A. Rowley, T. D. Ryba, B. C. Suh, J. Wei, D. W. Young, L. B. Akella, N. T. Ross, Y. L. Zhang, D. M. Fass, S. A. Reis, W. N. Zhao, S. J. Haggarty, M. Palmer, M. A. Foley, *J. Am. Chem. Soc.* **2010**, *132*, 16962–16976; c) X. Yang, L. P. Beroske, J. Kemmink, D. T. S. Rijkers, R. M. J. Liskamp, *Tetrahedron Lett.* **2017**, *58*, 4542–4546; d) I. Arora, V. K. Kashyap, A. K. Singh, A. Dasgupta, B. Kumar, A. K. Shaw, *Org. Biomol. Chem.* **2014**, *12*, 6855–6868; e) A. R. Kelly, J. Wei, S. Kesavan, J. C. Marié, N. Windmon, D. W. Young, L. A. Marcaurelle, *Org. Lett.* **2009**, *11*, 2257–2260.

[5] For selected reports on metal-free intramolecular azide-alkyne click reaction, see: a) D. K. Mohapatra, P. K. Maity, R. G. Gonnade, M. S. Chorghade, M. K. Gurjar, *Synlett.* **2007**, 1893–1896; b) V. Declerck, L. Toupet, J. Martinez, F. Lamaty, *J. Org. Chem.* **2009**, *74*, 2004–2007; c) K. B. Mishra, V. K. Tiwari, *J. Org. Chem.* **2014**, *79*, 5752–5762; d) J. K. Pokorski, L. M. M. Jenkins, H. Feng, S. R. Durell, Y. Bai, D. H. Appella, *Org. Lett.* **2007**, *9*, 2381–2383; e) F. Fang, M. Vogel, J. V. Hines, S. C. Bergmeier, *Org. Biomol. Chem.* **2012**, *10*, 3080–3091; f) R. A. Brawn, W. Morgan, J. T. Lowe, J. S. Panek, *Org. Lett.* **2010**, *12*, 336–339; g) V. Sovera, A. Bellomo, D. Gonzalez, *Tetrahedron Lett.* **2011**, *52*, 430–433; h) R. Li, D. J. Jansen, A. Datta, *Org. Biomol. Chem.* **2009**, *7*, 1921–1930; i) V. Sovera, L. Suescun, A. Bellomo, D. Gonzalez, *Eur. J. Org. Chem.* **2017**, 3912–3916.

[6] For selected reviews, see: a) S. Ma, *Acc. Chem. Res.* **2003**, *36*, 701–712; b) E. M. Beccalli, G. Broggini, M. S. Christodoulou, S. Giofrè, *Adv. Organomet. Chem.* **2018**, *69*, 1–71; c) T. Lechel, F. Pfrengle, H. U. Reissig, R. Zimmer, *ChemCatChem* **2013**, *5*, 2100–2130; d) S. Ma, *Top. Organomet. Chem.* **2005**, *14*, 183–210; e) A. Basso, L. Moni, R. Riva, *Multicomponent Reactions under Oxidative Conditions. In Multicomponent Reactions in Organic Synthesis*; J. Zhu, Q. Wang, M.-X. Wang, Eds.; Wiley-VCH: Weinheim, Germany, **2015**, pp 265–297.

[7] For selected reports on alcohol as nucleophiles in coupling cyclization of allenes, see: a) R. D. Walkup, L. Guan, M. D. Mosher, S. W. Kim, Y. S. Kim, *Synlett.* **1993**, 88–90; b) R. D. Walkup, L. Guan, Y. S. Kim, S. W. Kim, *Tetrahedron Lett.* **1995**, *36*, 3805–3808; c) S. Zhao, S. Ma, *J. Am. Chem. Soc.* **1999**, *121*, 7943–7944; d) W. Gao, S. Ma, *Synlett.* **2002**, 65–68; e) H. Ohno, H. Hamaguchi, M. Ohata, S. Kosaka, T. Tanaka, *J. Am. Chem. Soc.* **2004**, *126*, 8744–8754; f) X. Xie, S. Ma, *Chem. Commun.* **2013**, *49*, 5693–5695.

[8] For selected reports on carboxylic acid as nucleophiles in coupling cyclization of allenes, see: a) Z. Shi, S. Ma, *J. Org. Chem.* **1998**, *63*, 6387–6389; b) D. Duan, Z. Shi, S. Ma, *Org. Lett.* **2000**, *2*, 1419–1422; c) D. Duan, Y. Wang, S. Ma, *J. Comb. Chem.* **2002**, *4*, 239–247; d) J. Ye, S. Li, S. Ma, *Org. Biomol. Chem.* **2013**, *11*, 5370–5373.

[9] For selected reports on amine as nucleophiles in coupling cyclization of allenes, see: a) D. Lathhury, P. Verno, T. Gallagher, *Tetrahedron Lett.* **1986**, *27*, 6009–6012; b) R. C. Larock, Y. He, W. W. Leong, X. Han, M. D. Refvik, J. M. Zenner, *J. Org. Chem.* **1998**, *63*, 2154–2160; c) R. Grigg, V. Savic, V. Sridharan, C. Terrier, *Tetrahedron.* **2002**, *58*, 8613–8620.

[10] For selected reports on amide as nucleophiles in coupling cyclization of allenes, see: a) B. Chen, N. Wang, W. Fan, S. Ma, *Org. Biomol. Chem.* **2012**, *10*, 8465–8470; b) H. Luo, Z. Yang, W. Lin, Y. Zheng, S. Ma, *Chem. Sci.* **2018**, *9*, 1964–1969.

[11] For selected reports on sulfamide as nucleophiles in coupling cyclization of allenes, see: a) S. Kang, K. Kim, *Org. Lett.* **2001**, *3*, 511–514; b) W. Gao, S. Ma, *Org. Lett.* **2002**, *4*, 2989–2992; c) H. Ohno, H. Hamaguchi, M. Ohata, S. Kosaka, T. Tanaka, *Heterocycles.* **2003**, *61*, 65–68; d) K. Inamoto, A. Yamamoto, K. Ohsawa, K. Hiroya, T. Sakamoto, *Chem. Pharm. Bull.* **2005**, *53*, 1502–1507; e) J. Hu, B. Kong, Y. Liu, B. Xu, Y. Zhao, P. Gong, *ChemCatChem.* **2017**, *9*, 403–406.

[12] For selected reports on malonate as nucleophiles in coupling cyclization of allenes, see: a) M. Ahmar, B. Cazes, J. Gore, *Tetrahedron Lett.* **1985**, *26*, 3795–3798; b) S. Ma, S. Zhao, *Org. Lett.* **2000**, *2*, 2495–2497; c) H. O. Chang, C. Y. Rhim, C. H. Song, J. H. Ryu, *Chem. Lett.* **2002**, 1140–1141; d) N. Jiao, S. Zhao, H. Hou, S. Ma, *J. Org. Chem.* **2002**, *67*, 2837–2847; e) X. Jiang, Q. Yang, Y. Yu, C. Fu, S. Ma, *Chem. Eur. J.* **2009**, *15*, 7283–7286; f) X. Huang, X. Jiang, C. Fu, S. Ma, *Adv. Synth. Catal.* **2013**, *355*, 3295–3303.

 © 2019 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2020, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201901284 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

[13] a) S. Zhao, S. Ma, *J. Am. Chem. Soc.* **2001**, *123*, 5578–5579; b) S. Chen, Z. Gao, H. Zhao, B. Li, *J. Org. Chem.* **2014**, *79*, 1481–1486; c) X. Wu, C. Wang, W. Zhao, H. Zhao, S. Chen, *Synthesis.* **2018**, *50*, 4097–4103; d) Y. Hou, Q. Shen, Z. Li, S. Chen, Y. Zhao, M. Qin, P. Gong, *Adv. Synth. Catal.* **2018**, *360*, 631–636.

[14] a) S. Husinec, M. Jadranin, R. Markovic, M. Petkovic, V. Savic, N. Todorovic, *Tetrahedron Lett.* **2010**, *51*, 4066–4068; b) S. Husinec, M. Petkovic, V. Savic, M. Simic, *Synthesis.* **2012**, *44*, 399–408.

[15] H. M. Chang, C. H. Cheng, *J. Chem. Soc. Perkin Trans. 1*, **2000**, 3799–3807.

[16] a) M. Gardiner, R. Grigg, V. Sridharan, N. Vieker, *Tetrahedron Lett.* **1998**, *39*, 435–438; b) M. Gardiner, R. Grigg, M. Kordes, V. Sridharan, N. Vieker, *Tetrahedron.* **2001**, *57*, 7729–7735; c) X. Gai, R. Grigg, S. Rajviroongit, S. Songarsa, V. Sridharan, *Tetrahedron Lett.* **2005**, *46*, 5899–5902. d) C. Shin, Y. Oh, J. Cha, A. Pae, H. Cho, Y. Cho, *Tetrahedron* **2007**, *63*, 2182–2190.

[17] Crystal data for compound *cis*-**5jf**: $C_{32}H_{35}BrN_4O_2S$; MW = 619.61; triclinic, space group P-1; final R indices [I > 2\s(I)], R1 = 0.0642, wR2 = 0.1553, a = 10.8205(9) Å, b = 12.5451 (9) Å, c = 13.0649 (10) Å; α = 64.821(7), β = 71.393 (7), γ = 80.005 (7); V = 1519.6(2) Å^3; T = 296 K; Z = 2; reflections collected/unique 10015/ 5548 (R_{int} = 0.0401); number of observations [> 2\s(I)] 3086, parameters: 392. CCDC- 1891189 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

[18] Crystal data for compound *E*-**5kf**: $C_{29}H_{29}BrN_4O_2S$; MW = 557.53; monoclinic, space group P-1 21/n 1; final R indices [I > 2\s(I)], R1 = 0.0594, wR2 = 0.1614; a = 9.8259 (7) Å, b = 12.7335(8) Å, c = 21.8701(16) Å; α = 90, β = 92.339(7), γ = 90; V = 2734.1(3) Å^3; T = 296 K; Z = 4; reflections collected/unique 10823/5002 (R_{int} = 0.0488); number of observations [> 2\s(I)] 2833, parameters: 336. CCDC-1891198 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

[19] Crystal data for compound *cis*-**5lf**: $C_{27}H_{25}BrN_4O_2S$; MW = 549.48; triclinic, space group P −1; final R indices [I > 2\s(I)], R1 = 0.0299, wR2 = 0.0740; a = 9.7188 (3) Å, b = 9.8127(3) Å, c = 14.5107(4) Å; α = 91.4350(10), β = 102.6310(10), γ = 90; V = 1207.05(6) Å^3; T = 296 K; Z = 2; reflections collected/unique 24261/5340 (R_{int} = 0.0306); number of observations [> 2 \s(I)] 4847, parameters: 318. CCDC-1962422 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

[20] Crystal data for compound **5mf**: $C_{31}H_{31}BrN_4O_2S$; MW = 603.57; triclinic, space group P-1; final R indices [I > 2 \s(I)], R1 = 0.0667, wR2 = 0.1581, R indices (all data) R1 = 0.1395, wR2 = 0.2000; a = 10.9947(8) Å, b = 11.5050(9) Å, c = 12.7934(10) Å; α = 64.527(3), β = 83.033(3), γ = 81.345(4); V = 1441.43(19) Å^3; T = 170 K; Z = 2; reflections collected/unique 32053/13817 (R_{int} = 0.0940); number of observations [> 2\s(I)] 7171, parameters: 353. CCDC- 1891200 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

[21] a) A. Gagneux, S. Winstein, W. G. Young, *J. Am. Chem. Soc.* **1960**, *82*, 5956–5957; b) C. A. VanderWerf, V. L. Heasley, *J. Org. Chem.* **1966**, *31*, 3534–3536; c) A. A. Ott, M. H. Packard, M. A. Ortuno, A. Johnson, V. P. Suding, C. J. Cramer, J. J. Topczewski, *J. Org. Chem.* **2018**, *83*, 8214–8224; d) H. Y. Kang, Y. K. Kang, *Bull. Korean Chem. Soc.* **2017**, *38*, 1419–1426.

[22] S. Bera, G. A. Panda, *Org. Biomol. Chem.* **2014**, *12*, 3976–3985.

[23] H. Ohno, T. Mizutani, Y. Kadoh, A. Aso, K. Miyamura, N. Fujii, T. Tanaka, *J. Org. Chem.* **2007**, *72*, 4378–4389.

[24] J. Park, S. Kim, S. Kim, S. Lee, Y. Chung, *Synlett* **2007**, 453–459

[25] C. Mukai, Y. Hara, Y. Miyashita, F. Inagaki, *J. Org. Chem.* **2007**, *72*, 4454–4461.

[26] a) X. Huang, T. Cao, Y. Han, X. Jiang, W. Lin, J. Zhang, S. Ma, *Chem. Commun.* **2015**, *51*, 6956–6959; b) X. Huang, C. Xue, C. Fu, S. Ma, *Org. Chem. Front.* **2015**, *2*, 1040–1044.

[27] X. Tang, C. Zhu, T. Cao, J. Kuang, W. Lin, S. Ni, J. Zhang, S. Ma, *Nat. Commun.* **2013**, *4*, 2450.

[28] Y. Xiang, Z. Li, L. Wang, Z. Yu, *J. Org. Chem.* **2018**, *83*, 7633–7647.

 © 2019 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2020, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201901284 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License